Computer-Aided Design 179 (2025) 103811

Contents lists available at ScienceDirect

Computer-Aided Design

journal homepage: www.elsevier.com/locate/cad

Research Paper

Two-Level High-Resolution Structural Topology Optimization with Equilibrated Cells

Rafael Merli [a,b,*], Antolín Martínez-Martínez [a], Juan José Ródenas [a], Marc Bosch-Galera [a], Enrique Nadal [a]

[a] *Departamento de Ingeniería Mecánica y de Materiales. Escuela Técnica Superior de Ingenieros Industriales, Universitat Politècnica de València. Camino de vera s/n, 46022 Valencia, Spain*

[b] *Centro Universitario EDEM - Escuela de Empresarios, Muelle de la Aduana s/n. La Marina de Valencia, 46024 Valencia, Spain*

ARTICLE INFO

Keywords:
Topology optimization
Two-level method
Traction field
Continuity
Additive manufacturing

ABSTRACT

In today's industry, the rapid evolution in the design and development of optimized mechanical components to meet customer requirements represents a significant challenge for companies. These companies seek efficient solutions to enhance their products in terms of stiffness and strength. One powerful approach is Topology Optimization, which aims to determine the optimal material distribution within a predefined domain to maximize the overall component's stiffness. Achieving high-resolution solutions is also crucial for accurately defining the final material distribution. While standard Topology Optimization tools can propose optimal solutions for entire components, they struggle with small-scale details (such as trabecular structures) due to prohibitive computational costs. To address this issue, our proposed approach introduces a two-level topology optimization methodology considering density-based techniques. The proposed methodology includes three steps: The first one subdivides the whole component in cells and generates a coarse optimized low-definition material distribution, assigning a different density to each cell. Since the output stresses from the coarse problem are not equilibrated into each cell, they must not be directly used in the fine level. Thus, the second step uses the equilibrating traction recovery approach to convert the cell nodal forces into equilibrated lateral tractions over the cell boundary. Finally, taking as input data the densities from the coarse optimization and imposing these lateral tractions as Neumann boundary conditions, each cell is optimized at fine level. The main goal of this work is to efficiently solve high-resolution topology optimization problems using a two-level mechanically-continuous method, which would be unaffordable with standard computing facilities and the current techniques.

1. Introduction

The optimization of mechanical components represents a major challenge in scientific and engineering fields. Before the huge development of computer science, structural optimization was the result of a long trial and error process. In fact, skills and intuition of engineers strongly conditioned the final designs. However, the increasing economical, industrial and social requirements need automated processes to develop optimized structures surpassing the speed of traditional procedures.

Numerical simulation methodologies can considerably help to take better decisions along the design phase of complex structures. These numerical methods, e.g. the Finite Element Method (FEM), allow us to evaluate accurate predictions of structural components subjected to external loads and to propose high-performance material distributions. For instance, the 3D shape of structural components is defined in [1] considering geometrical parameters (like diameters, section's heights, etc.) and a shape optimization algorithm is used to evaluate the values of the parameters that minimize the mass of the components considering stress constraints to avoid failure. In these shape optimization problems the connectivity of the structure, i.e. its topology, is always maintained.

One of the early works aimed to obtain the optimal distribution of material into a design domain [2] introduced the theory of numerical homogenization to evaluate the mechanical properties of predefined square cells with rectangular holes. Alternatively, the well-known Solid Isotropic Material with Penalization (SIMP) [3–7] can yield reasonable results to solve very large models using a much simpler formulation.

* Correspondence to: Universitat Politècnica de València, Departamento de Ingeniería Mecánica y de Materiales, Camino de vera s/n, 46022 Valencia, Spain.
E-mail address: ramergis@upvnet.upv.es (R. Merli).

https://doi.org/10.1016/j.cad.2024.103811
Received 14 May 2024; Received in revised form 3 October 2024; Accepted 7 October 2024
Available online 17 October 2024

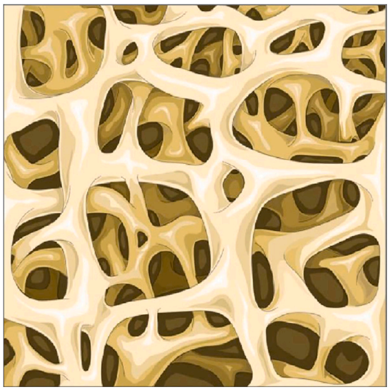


Fig. 1. Trabecular structure of human bone.
Source: https://commons.wikimedia.org/wiki.

Other methods, such as the evolutionary [8] or the level set methods [9,10] have shown satisfactory results. More detailed classifications of topology optimization methods can be found in [11,12].

Since many structures optimized by nature present trabecular topologies (e.g. bone tissues, see Fig. 1) working properly with lightweight material distributions, it would be highly desirable to reproduce this trabecularity with high resolution as the output of topology optimization problems. In fact, these trabecular structures span multiple length scales and, according to [12], have proven their versatility to withstand load variations from those set in the initial design. Moreover, they are also more tolerant to bar instabilities since local buckling can be controlled directly by the cell size and the minimum bar thickness inside each cell.

As can be commonly accepted, regarding single-scale (or full-scale) models, an increase in mesh discretization produces results with high definition and may be used to cover the multiple length scales present in real materials. However, this involves sometimes a huge computational cost [13]. To alleviate this expense, the so-called multiresolution topology optimization method [14] applies the SIMP model to single-scale topology optimization, obtaining a density distribution into each FE from the design variables via a projection function, which allows to evaluate the densities in the Gauss points in order to integrate the element stiffness matrix. Finally, these projected densities are represented in a finer mesh to potentially achieve a high-resolution design. Even if an increase in the sharpness is obtained, the relative size between the FE and the density meshes does not reach one order of magnitude, which is not enough to produce scale separation. Besides, this approach is not specifically devised to generate trabecular structures.

On the other hand, multi-scale modelling assumes separation between length scales using the theory of numerical homogenization to evaluate the effective mechanical properties of each macroscale cell from the microstructure details. As seen in [15], for problems with many design variables, splitting the analysis into macro and micro problems enables the improvement of the computational performance via parallelization techniques. This numerical advantage along with the additive manufacturability of the obtained designs have reinforced the development of multi-scale approaches, even if suboptimal solutions were yielded.

Included in the multi-scale methods, laminates rank$-N$ introduce microstructure parameterizations which result in near-optimal solutions. The total number of parameters N needed to determine the microstructure depends on the dimension of the problem and the applied load cases. In fact, since they are enough to define the homogenized constitutive model, [16] established that rank-2 laminates for 2D domains under a single load case and rank-3 laminates under multiple load cases are able to provide optimal solutions. As examples, [17] defined a rank-2 laminate including a rotation angle to align the microstructure with the principal stress directions, [18] extended this work to 3D problems with multiple load cases and more recently, [19] employed rank-3 laminates to systematically create almost optimal microstructures. Despite of the interesting advantages of these laminates, they are not conveniently aimed to reproduce trabecular structures due to the restricted configuration of the microstructure parameterization.

Following a similar approach, the microstructure can be defined by several prescribed unit cells as a function of a few size parameters, generating periodic lattice microstructures by repetition of these cells solving the optimization problem at the coarse scale. Thus, [20] used predefined lattice structures (square frame and frame with crossed trusses) into each element and related the density of the cell with the characteristic thickness. Accordingly, [21] employed more sophisticated unit cells and carried out the concurrent topology optimization at macro and micro scales. Later, [22] studied the stiffness and buckling response of two different microstructures. On the other hand, [23] used periodic cells to obtain graded microstructures, which can be achieved using additive manufacturing. Overall, the more parameters are included in the definition of each unit cell, the better adaptation of the microstructural anisotropy to the stress directions is yielded. This insight was first outlined in the seminal work [2], releasing the rotation of the unit cell.

Although lattice structures are usually defined to keep the geometrical continuity between neighbouring cells, the obtained solution are clearly suboptimal because the initial restraints imposed into each cell prevent them from being completely optimized. As an alternative to these lattice structures, other works [24] propose a new strategy based on generating a database of fine-level optimized solutions, each of them corresponding to a single macroscopic stress state. Finally, the union of all these cells would provide the optimized solution of the entire domain. However, the solution obtained by this method is not structurally viable because the continuity between cells has not been taken into account.

As shown, the challenging issue of the continuity between completely optimized neighbouring cells in multiscale problems is a challenge. In this sense, several works propose alternatives to face this issue. For instance, [25] addressed the microstructural optimization of connective cells in 2D domains. Namely, periodicity of cells is assumed in one direction whereas a smooth linear variation of density is adopted in the other. Hence, the concurrent topology optimization of each individual cell and each couple of neighbouring cells in the non-periodic direction is rendered to ensure the connectivity with a graded variation in the amount of material. The compatibility in the direction of periodicity is established via a rotational symmetry constraint into each cell.

Moreover, [26] performed the concurrent two-scale topology optimization in 2D and 3D domains, but setting a predefined adjustable region of connectivity between neighbouring cells, where their density is forced to be the same as an additional constraint in the optimization problem. Thus, the obtained material distributions are clearly manufacturable, but still suboptimal. The design space is restrained since only a reduced number of lattices are considered, with inner members into each unit cell unable to be aligned with the principal stress directions, thus leading to suboptimal solutions.

One of the main advantages of the proposed methodology against previous works is to provide trabecular topology optimized structures with explicit consideration of the mechanical continuity at a low computational cost, based on the use of standard computational resources. In that sense, we propose a two-level SIMP-based topology optimization method. In the coarse level, the SIMP process [3] with a low penalty value provides a diffuse material distribution which determines the amount of material that has to be considered into each of these coarse elements, also called cells. Then, in the fine level another SIMP-based topology optimization process is performed using a fine mesh into each element of the coarse level with a high (normally $p = 3$) penalty value, thus providing a sharp representation of the topology at the cell level.

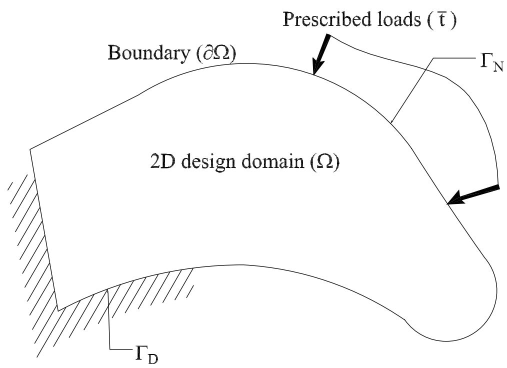


Fig. 2. 2D design problem.

In order to connect both levels, a post-processed traction field over each cell boundary is performed via the element-wise traction equilibrium method depicted in [27]. This procedure was initially used in the field of error bounding to postprocess the discontinuous finite element stress field, in order to produce a statically admissible solution defined by equilibrated tractions acting on the boundaries of the elements. In our proposed methodology, this statically admissible solution will play two roles. In the first place, the equilibrated tractions will be used as equilibrated boundary conditions for the fine-level optimization problems and will allow to solve them independently from each other. In the second place, the equilibrated tractions between adjacent cells will smoothly impose the inter-cell structural continuity. We have to emphasize that the resulting space of trabecular solutions is richer than the corresponding ones using the previous approaches described in the literature since no topological constraints are considered.

Although the proposed methodology allows us to obtain a high-resolution topology using a two-level approach, it cannot be considered as a multiscale approach since no strict scale separation is performed.

The paper is organized as follows: Section 2 summarizes the basics of 2D optimization problems employed herein. In Section 3 the proposed two-level optimization methodology and its numerical implementation are presented. Sample problems and their numerical results compared with some published references are presented in Section 4. Finally, concluding remarks are addressed in Section 5.

2. Main methodologies

2.1. Two-dimensional linear elasticity problem

For the sake of clarity, the general 2D linear elasticity problem and the notation used hereafter are described next.

We seek a displacement field $\mathbf{u}$ and a Cauchy stress field $\boldsymbol{\sigma}$ defined over the domain Ω (Fig. 2) verifying:

- Static admissibility, involving internal equilibrium and Neumann boundary conditions (BC)

$$\mathbf{L}^T\boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \qquad \text{in } \Omega \tag{1a}$$

$$\mathbf{G}\boldsymbol{\sigma} = \bar{\mathbf{t}} \qquad \text{in } \Gamma_N, \tag{1b}$$

where $\mathbf{L}$ is the differential operator defined by:

$$\mathbf{L} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \tag{2}$$

and $\mathbf{G}$ is the operator which projects the stress field $\boldsymbol{\sigma} = \{\sigma_x, \sigma_y, \tau_{xy}\}^T$ into tractions over the boundary. Considering the external unit vector $\mathbf{n} = \{n_x, n_y\}^T$ normal to the boundary $\partial\Omega$ with Neumann BC Γ_N, then:

$$\mathbf{G} = \begin{bmatrix} n_x & 0 & n_y \\ 0 & n_y & n_x \end{bmatrix}. \tag{3}$$

- Dirichlet BC on $\Gamma_D \subset \partial\Omega$ (note that $\Gamma_D \cap \Gamma_N = \emptyset$)

$$\mathbf{u} = \mathbf{0}. \tag{4}$$

- Constitutive relations:

$$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon}(\mathbf{u}), \tag{5}$$

where $\boldsymbol{\varepsilon}(\mathbf{u}) = \mathbf{L}\mathbf{u} = \{\varepsilon_x, \varepsilon_y, \gamma_{xy}\}^T$ and $\mathbf{D}$ contains the mechanical parameters of the elastic isotropic material.

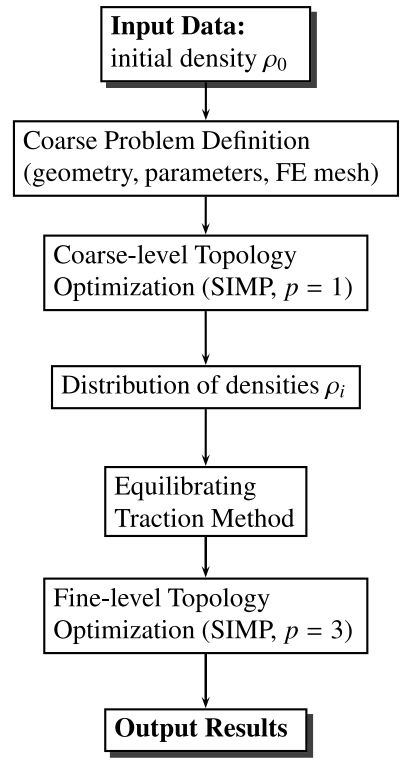


Fig. 3. General two-level topology optimization procedure.

Alternatively, the problem above can be expressed in variational form by defining:

$$a(\mathbf{u},\mathbf{v}) = \int_\Omega \boldsymbol{\varepsilon}(\mathbf{u})^T \mathbf{D}\boldsymbol{\varepsilon}(\mathbf{v}) d\Omega \qquad l(\mathbf{v}) = \int_\Omega \mathbf{b}^T\mathbf{v} d\Omega + \int_{\Gamma_N} \bar{\mathbf{t}}^T \mathbf{v} d\Gamma_N, \tag{6}$$

thereby, we should:

$$\text{Find } \{\mathbf{u} \in (V \cup \{\mathbf{w}\})/a(\mathbf{u},\mathbf{v}) = l(\mathbf{v})\} \qquad \forall v \in V, \tag{7}$$

$$\text{where } V = \{\mathbf{v} \in C^1(\Omega)/\mathbf{v}|_{\Gamma_D} = \mathbf{0}\} \tag{8}$$

and $\mathbf{w}$ is a particular displacement field satisfying the Dirichlet BC (4).

2.2. Topology optimization problem

Following [4], the SIMP method can be adopted to solve the topology optimization problem for minimum compliance, which takes the form:

$$\min \{a(\rho; \mathbf{u}, \mathbf{u})\}, \qquad \forall \mathbf{u} \in (V \cup \{\mathbf{w}\}), \tag{9}$$

subjected to:

$$l(\mathbf{u}) = a(\rho; \mathbf{u}, \mathbf{u}) \tag{10a}$$

$$\int_V \rho(\mathbf{x})dV = V_0,\ \forall \mathbf{x} \in \Omega, \tag{10b}$$

$$\rho \geq \rho_{min} \tag{10c}$$

$$\rho \leq 1 \tag{10d}$$

where:

$\mathbf{u}$	unknown displacements
V_0	initial volume value to be distributed in the domain Ω
$\rho(\mathbf{x})$	relative density function at position $\mathbf{x}$
$\rho_{min} = 10^{-3}$	minimum relative density threshold, used to avoid singularities in the equilibrium problem.

The constitutive matrix $\mathbf{D}$ is redefined with respect to that of a reference material $\mathbf{D}_0$, as follows:

$$\mathbf{D}(\mathbf{x}) = \rho^p(\mathbf{x})\mathbf{D}_0, \tag{11}$$

where $p \geq 1$ is a penalty parameter which penalizes intermediate densities (usually $p = 3$). Therefore, the functional $a(\mathbf{u}, \mathbf{v})$ in (6) becomes

$$a(\rho; \mathbf{u}, \mathbf{u}) = \int_\Omega \rho^p \boldsymbol{\varepsilon}(\mathbf{u})^T \mathbf{D}_0 \boldsymbol{\varepsilon}(\mathbf{u}) d\Omega. \tag{12}$$

2.2.1. Optimality criteria

Following [4, Section 1.2.1], Lagrange multipliers to the equivalent Lagrange function $\mathcal{L}$ of the optimization problem given by Eq. (9) subjected to constraints (10) can be applied. Likewise, considering intermediate densities, a new design variable to enable the density updating is defined into each iteration k as

$$B_k = \frac{1}{\Lambda_k} p\rho(\mathbf{x})^{p-1} \boldsymbol{\varepsilon}^T(\mathbf{u}_k)\mathbf{D}_0\boldsymbol{\varepsilon}(\mathbf{u}_k). \tag{13}$$

where Λ_k is obtained through the bisection method into each iteration k. Thus, the following fix-point-like density updating for each element is adopted [4]:

$$\rho^e_{k+1} = \begin{cases} \max\{(1-\zeta)\rho^e_k, \rho_{min}\} & \text{if } \rho^e_k B^\eta_k \leq \max\{(1-\zeta)\rho^e_k, \rho_{min}\} \\ \rho^e_k B^\eta_k & \text{if } \max\{(1-\zeta)\rho^e_k, \rho_{min}\} \\ & \quad \leq \rho_k B^\eta_k \leq \min\{(1+\zeta)\rho^e_k, 1\} \\ \min\{(1+\zeta)\rho^e_k, 1\} & \text{if } \rho^e_k B^\eta_k \geq \min\{(1+\zeta)\rho^e_k, 1\}, \end{cases} \tag{14}$$

being $\zeta = 0.2$ and $\eta = 0.5$ the standard values of the parameters used to control the process. From (13), a local optimum is reached if $B_k = 1$ for densities $\rho_{min} < \rho < 1$. In addition, the updating scheme (14) adds material to areas where $B_k > 1$, related to high variations of strain energy density in the numerator of (13). Otherwise, material is removed from areas where $B_k < 1$, namely, with low variations of strain energy density.

3. The proposed two-level optimization technique

3.1. General procedure

In order to provide an overview of the proposed method (see Fig. 3), the main steps of the two-level optimization technique are outlined next. Firstly, using a coarse discretization, a coarse-level topology optimization problem is solved via the SIMP method (Fig. 4(a)) considering a penalty value $p = 1$ (see (11)), as the aim of this first step is to obtain a description of the mass distribution along the problem domain defined as a coarse representation of the relative density field. Secondly, a traction equilibrating method [27] is applied (see Section 3.2 and Fig. 4(b)). This method uses the converged solution of the coarse-level problem, whose stress field is discontinuous, to obtain a set of equilibrated tractions on the boundary of the elements of this coarse mesh. The domain defined by each of the elements of the coarse mesh will define a cell, that discretized with a fine mesh, will be used in the last step to run the topology optimization process at the fine level (see Section 3.3 and Fig. 4(c)). The equilibrated tractions will be used as Neumann boundary conditions, converted into nodal forces acting on the external nodes of the fine mesh for each cell. The fact that the tractions between adjacent cells are equilibrated will lead to mechanical continuity between cells, which represents a relevant improvement to previous works, like [24], as it will provide structurally viable solutions.

Note that if the fine-level elements were infinitely small, the final topology in the inner region adjacent to the cell contour would be driven by the local tractions applied to the contour. Since these tractions are equilibrated between cells, the local topology at each side of the boundary between adjacent cells will tend to be the same when refining the mesh. Apart from the lack of full mechanical continuity induced by the use of finite-size elements, there is another source of discontinuity between adjacent cells that comes from the different values of relative densities evaluated at each of the cells of the coarse discretization, that is, from the discontinuous density field evaluated in the coarse problem. This effect depends on the size of the elements of the coarse mesh, but it could be smoothed by using h-adaptive techniques [28].

As in several previous works, e.g. in [29,30] with microstructure approaches and [31] with the SIMP method, the initial volume fraction ρ_0 is adopted as a constant value for each simulation. When the SIMP method is applied, we usually consider $p = 3$ to penalize intermediate density values, as has been done herein at the fine level. However, a value of $p = 1$, that does not penalize intermediate densities, is adopted at the coarse level to obtain a low-definition description of the optimal material distribution.

It should be taken into account that all calculations through the FEM at both levels (coarse and fine) are performed using Cartesian grids [32] with four-node square bilinear elements. For implementation simplicity, only domains which can be fitted by Cartesian grids are included. However, the methodology presented in this paper could be extended to other geometries if immersed boundary methodologies, like the Cartesian grid FEM (cgFEM) [1] or the Finite Cell method [33], were considered.

Regarding the computational efficiency, the computational cost is reduced by introducing the proposed two-level scheme in comparison with a single-level approach with the same mesh resolution. Consider that the coarse problem is formed by N cells and each fine problem by n elements. Taking into account that the main part of the computational cost is devoted to solve the FE linear system of equations, the computational cost of the proposed two-level approach is $\mathcal{O}(N(n^p))$ with $p > 1$ the scaling parameter of the solver used, typically $p = 2$ or $p = 3$ (see [34]). On the contrary, when a single-level approach is used, the computational cost is of order $\mathcal{O}((Nn)^p)$. Therefore, the saving ratio is of order $\mathcal{O}(N^{(p-1)})$.

Additionally, besides the numerical independency among cells, their optimization in the fine level might be solved in parallel. On the other hand, the fact that all cells have the same square shape would simplify the use of machine learning strategies to achieve significant improvements in the computational performance with respect to the full-scale approach for similar resolution.

3.2. The traction equilibrating method

As mentioned in Section 1, the loads for each of the fine-level topology optimization problem are obtained from the coarse problem solution. The more straightforward way to transfer these loads is by using the stresses obtained from the coarse FEM analysis, as considered in [24]. However, that method presents some difficulties as the FE stresses are discontinuous between elements and because the FE stress field in the element is not equilibrated.

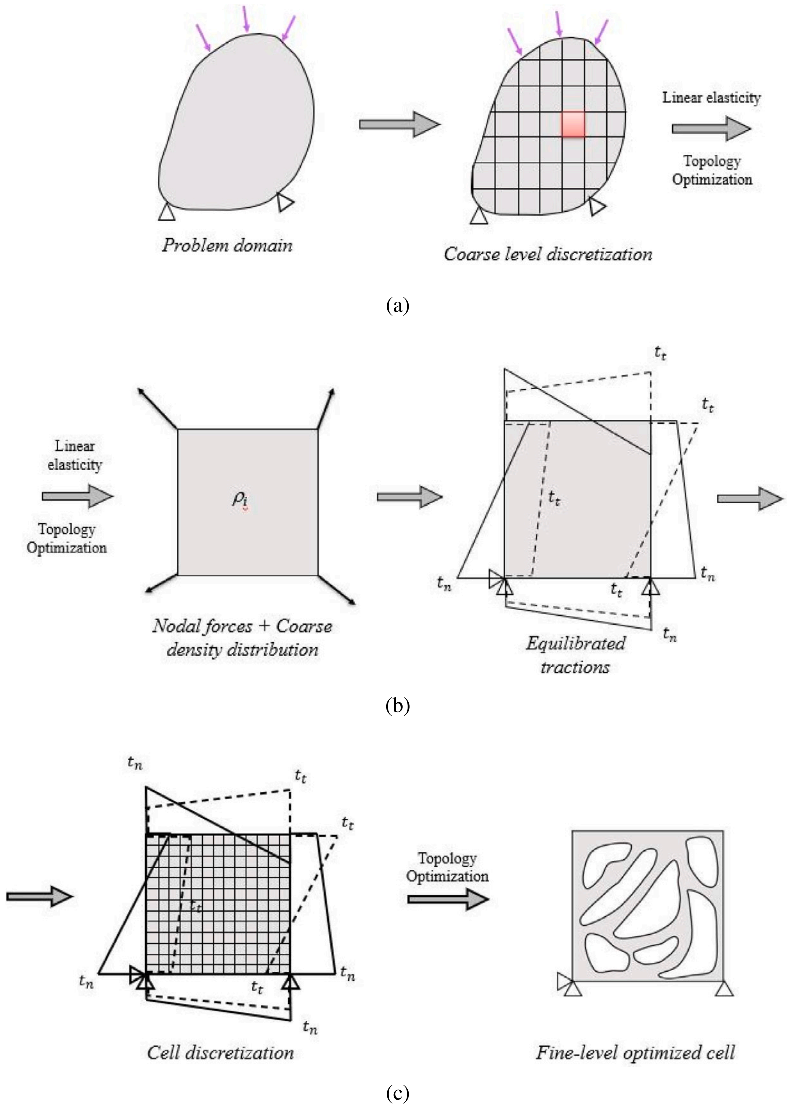


Fig. 4. General two-level topology optimization graphical scheme.

In this work the traction equilibrating method introduced in [27] is applied to obtain a continuous traction field along the edges between elements at the coarse level, which is properly converted into nodal forces using the standard shape functions acting on the external nodes of the fine mesh over each cell. In addition, we must impose sufficient constraints to prevent the rigid body motion of each coarse element (see Fig. 4(c)), so that the topology optimization problem at fine level can be solved. Following [27], the applied method operates in three main steps (Fig. 5): from the FE solution of the coarse problem, the nodal forces $\hat{\mathbf{F}}_k^E, \hat{\mathbf{F}}_l^E, \hat{\mathbf{F}}_m^E, \hat{\mathbf{F}}_n^E$ are obtained for each coarse element (Fig. 5(a)). Each of them (for instance, $\hat{F}_n^E$) is then distributed into two side forces $\hat{\mathbf{P}}_{in}^E, \hat{\mathbf{P}}_{jn}^E$ (Fig. 5(b)) in such a way the element satisfies equilibrium with its neighbours. Finally, at each side i of the element a linear traction distribution $\hat{\mathbf{t}}_i$, statically equivalent to $\hat{\mathbf{P}}_{il}$, $\hat{\mathbf{P}}_{im}$, is determined (Fig. 5(c)). Note that each $\hat{\mathbf{t}}$ will have two components, although, for clarity purposes, only normal stresses have been represented in Fig. 5(c).

Since the calculation details of the side forces $\hat{\mathbf{P}}_n$ depend on the number of elements surrounding each specific node into the Cartesian grid, an algorithm was designed to classify each into three main groups: internal nodes, nodes with Dirichlet BC and nodes with Neumann BC.

Once the node connectivity is known, the main equations used to solve the side forces $\hat{\mathbf{P}}_{in}$ over a coarse-level element are outlined below (see [27] for details).

3.2.1. Internal nodes

Nodes without any external force or reaction directly acting on them and surrounded by four elements are included herein. Volume forces are not taken into account throughout this work. For each element, the nodal forces $\hat{F}^e$ are obtained from the FE solution using the following expression:

$$\hat{\mathbf{F}}^e = \mathbf{K}^e \mathbf{u}^e, \tag{15}$$

where $\mathbf{K}^e$ is the stiffness matrix and $\mathbf{u}^e$ are the nodal displacements of element e.

Consider elements E, M, B, A connected to node n as shown in Fig. 5(d). As nodes are in equilibrium, if we use a Maxwell diagram to represent the nodal forces from these elements $(\hat{\mathbf{F}}_n^E, \hat{\mathbf{F}}_n^M, \hat{\mathbf{F}}_n^B, \hat{\mathbf{F}}_n^A)$ acting on n, we obtain a closed quadrilateral polygon (see Fig. 6(a)). In addition, the action–reaction principle, $\hat{\mathbf{P}}_{in}^E = -\hat{\mathbf{P}}_{in}^M$, and the equilibrium in each corner, $\hat{\mathbf{P}}_{in}^E + \hat{\mathbf{P}}_{jn}^E = \hat{\mathbf{F}}_n^E$ (Fig. 5(b)) are kept for the side nodal forces, as graphically shown in Fig. 6(a), where point P is called *pole*. The force splitting $\hat{\mathbf{P}}_{in}^E + \hat{\mathbf{P}}_{jn}^E = \hat{\mathbf{F}}_n^E$ is not unique, hence, we can select one of the possible solutions simply selecting the location of the *pole* P. Although this location could be arbitrarily selected, following [27], in order to avoid excessively large side nodal forces and to avoid a large computational cost, the *pole* has been chosen at the geometrical mass centre of the polygon as

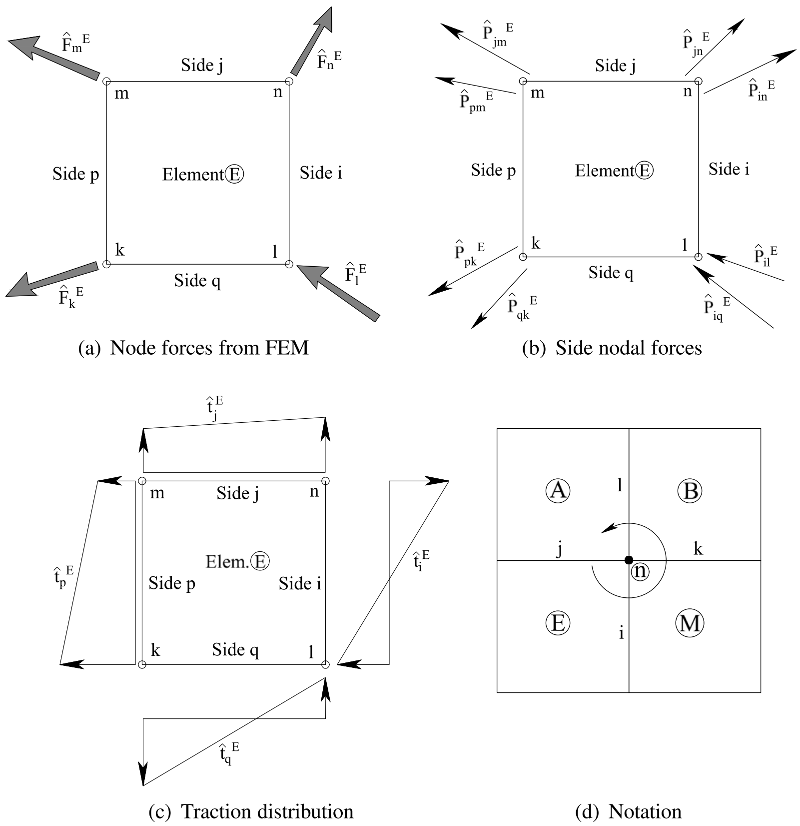


Fig. 5. Steps and notation in the traction equilibrating method.

$$\hat{\mathbf{P}}_{in}^{E} = \mathbf{r}_G, \tag{16}$$

where the origin of coordinates of the polygon has been taken at vertex i in Fig. 6(a), which includes the Maxwell polygon for the sake of completeness. In any case, other approaches can be followed as described in [27].

Rearranging the mentioned equations in matrix form, the following system is derived:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \hat{\mathbf{P}}_{in}^{E} \\ \hat{\mathbf{P}}_{jn}^{E} \\ \hat{\mathbf{P}}_{jn}^{A} \\ \hat{\mathbf{P}}_{ln}^{A} \\ \hat{\mathbf{P}}_{ln}^{B} \\ \hat{\mathbf{P}}_{kn}^{B} \\ \hat{\mathbf{P}}_{kn}^{M} \\ \hat{\mathbf{P}}_{in}^{M} \\ \lambda \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{F}}_{n}^{E} \\ \hat{\mathbf{P}}_{n}^{A} \\ \hat{\mathbf{P}}_{n}^{B} \\ \hat{\mathbf{P}}_{n}^{M} \\ 0 \\ 0 \\ 0 \\ 0 \\ \mathbf{r}_G \end{Bmatrix}, \tag{17}$$

where λ is a force residual parameter introduced to make the system in (17) well-posed and will end up to be zero. It should be noted that each of the forces $\hat{\mathbf{P}}_{in}$, $\hat{\mathbf{F}}_n$ and λ have two components (x and y directions). Thus, the system (17) is sized 18×18 and its solution will provide the eight side nodal forces $\hat{\mathbf{P}}^{E}$ for each element (two per side).

A simple strategy has been adopted to obtain the geometrical mass centre of the Maxwell polygon. Firstly, the quadrilateral is divided into two triangles along its diagonal $i-k$ (Fig. 6(b)) and secondly, the areas and centre of masses of both triangles are obtained by using two edge vectors taken in an anticlockwise sense (Fig. 6(c)), obtaining (e.g. for ijk) their properties from:

$$A_1 = \frac{1}{2}(\hat{\mathbf{F}}_n^{E} \times \hat{\mathbf{F}}_n^{M})_z \tag{18a}$$

$$\mathbf{r}_{G_1} = \hat{\mathbf{F}}_n^{E} + \frac{2}{3}\boldsymbol{\mu}, \tag{18b}$$

where $\boldsymbol{\mu}$ is the median vector from vertex j and $(\,)_z$ indicates the third component of the cross product.

Eqs. (18) can be adapted to triangle ikl. The final geometrical mass centre of the quadrilateral is obtained as

$$\mathbf{r}_G = \frac{\mathbf{r}_{G_1} A_1 + \mathbf{r}_{G_2} A_2}{A_1 + A_2}. \tag{19}$$

Note that the possible *concavity* in one vertex of the quadrilateral does not modify Eq. (19), even if the sense of the cross product (18a) were inverted. Nevertheless, if this *concavity* leads to intersect two opposite sides of the polygon (see Fig. 6(d) as an example), Eq. (19) should be replaced by

$$\mathbf{r}_G = \frac{\mathbf{r}_{G_1}|A_1| + \mathbf{r}_{G_2}|A_2| - 2\mathbf{r}_{G_3}|A_3|}{|A_1| + |A_2| - 2|A_3|} \tag{20}$$

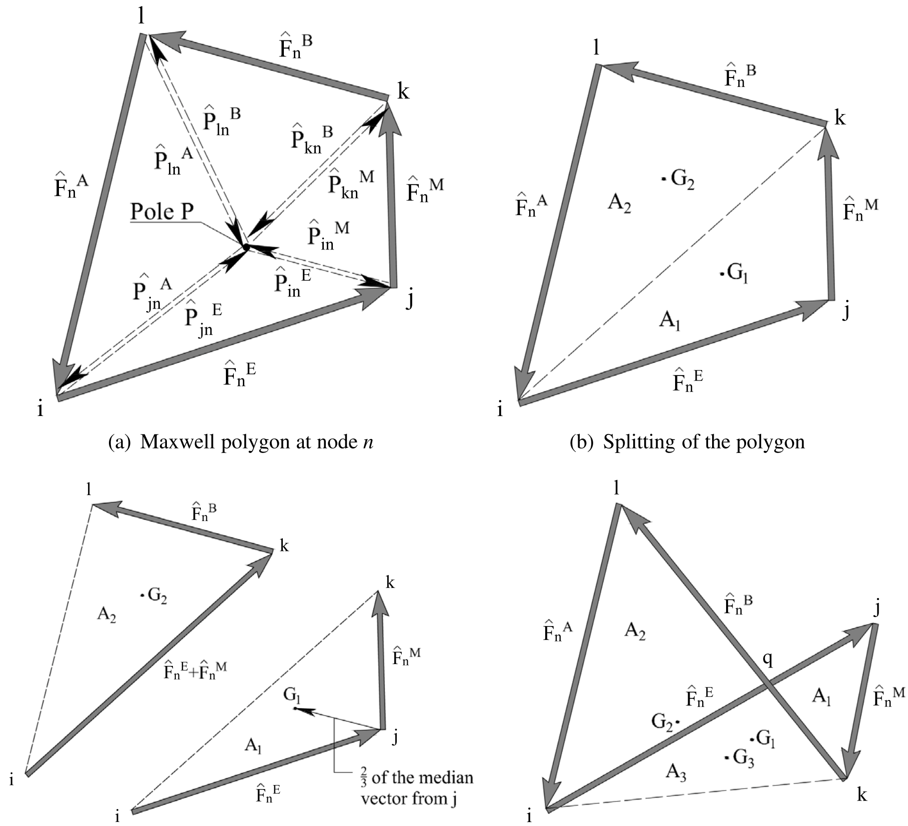
in order to remove overlapping areas in triangle ikq and the subsequent inconsistency of equilibrium.

The following equation is used in the FEM to evaluate equivalent nodal forces $\hat{\mathbf{P}}_{in}^{E}$ from tractions $\hat{\mathbf{t}}_i^{E}$ applied on element sides:

$$\int_i \phi_n^E \hat{\mathbf{t}}_i^E \, dS = \int_i \phi_n^E (\phi_n^E \hat{\mathbf{t}}_{in}^E + \phi_m^E \hat{\mathbf{t}}_{im}^E) dS = \hat{\mathbf{P}}_{in}^E, \tag{21}$$

where ϕ_n^E, ϕ_m^E represent the standard shape functions along side i. In our case, $\hat{\mathbf{P}}_{in}^E$ is obtained from (17), hence, from (21) we can evaluate the nodal values of $\hat{\mathbf{t}}_{in}^E$, $\hat{\mathbf{t}}_{im}^E$ that define the linear tractions on side i solving the following system of equations:

$$\begin{bmatrix} \int_i (\phi_n^E)^2 dS & \int_i (\phi_n^E \phi_m^E) dS \\ \int_i (\phi_m^E \phi_n^E) dS & \int_i (\phi_m^E)^2 dS \end{bmatrix} \begin{Bmatrix} \hat{\mathbf{t}}_{in}^E \\ \hat{\mathbf{t}}_{im}^E \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{P}}_{in}^E \\ \hat{\mathbf{P}}_{im}^E \end{Bmatrix}, \tag{22}$$



(a) Maxwell polygon at node n

(b) Splitting of the polygon

(c) Strategy to calculate each G_i

(d) Example of intersection between opposite sides

Fig. 6. Stages to calculate the centre of masses of Maxwell polygon.

The resulting values of system (22) will be used afterwards in the fine-level topology optimization.

Sections 3.2.2 and 3.2.3 will outline the main equations used for nodes under BC. Further details can be found in [27].

3.2.2. Nodes with Dirichlet BC

In the case of 2D Cartesian grids these nodes will be surrounded either by one or by two elements.

Standard case. The first case is defined by two elements surrounding one node on Γ_D. In this case, the reaction force will be divided in two components $\mathbf{R}_n^E, \mathbf{R}_n^M$ that are calculated together with the interface forces $\hat{\mathbf{P}}_{in}^E$, $\hat{\mathbf{P}}_{in}^M$. Thus, the Maxwell diagram is a triangle and the pole is located at the centre of masses through Eq. (16), which is equivalent in this case to

$$\hat{\mathbf{P}}_{in}^E = \mathbf{R}_n^E - \mathbf{R}_n^M. \tag{23}$$

Rearranging equations in matrix form, the system that represents the equilibrium and the action–reaction principle is written as

$$\begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{0} & -\mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{0} & \mathbf{1} & \mathbf{0} & -\mathbf{1} \\ -\mathbf{1} & \mathbf{1} & \mathbf{0} & -\mathbf{1} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \hat{\mathbf{P}}_{in}^E \\ \mathbf{R}_n^E \\ \hat{\mathbf{P}}_{in}^M \\ \mathbf{R}_n^M \\ \lambda \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{F}}_n^E \\ \hat{\mathbf{F}}_n^M \\ \mathbf{R}_n \\ \mathbf{0} \\ \mathbf{0} \end{Bmatrix}. \tag{24}$$

Outer corner node. Assuming a reaction $\mathbf{R}_n$ is applied on the node, its module is equal to the node force, but with opposite sign. This means that there is no force left to add, as this node will be equilibrated. Therefore the forces applied on this node are obtained straightaway.

3.2.3. Nodes with Neumann BC

The analysis of nodes under Neumann BC is quite similar to that in Section 3.2.2. However, the external forces are known since our developed code properly distributes the tractions on nodes in Γ_N, previous to the FEM analysis of the optimization process. The following three cases are studied:

Standard case The first case is defined as a boundary node surrounded by two elements under Neumann BC, the external forces $\mathbf{T}_n^E, \mathbf{T}_n^M$ applied in each element connected with the same node are calculated previously to the FEM optimization analysis. Now, the Maxwell diagram is a quadrilateral with the *pole* located at one vertex (see Fig. 6 in [27]). The corresponding system of equations takes the form:

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \hat{\mathbf{P}}_{in}^E \\ \hat{\mathbf{P}}_{in}^M \\ \lambda \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{F}}_n^E - \mathbf{T}_n^E \\ \hat{\mathbf{F}}_n^M - \mathbf{T}_n^M \\ \mathbf{0} \end{Bmatrix}. \tag{25}$$

Obviously, when no external forces are applied, Eq. (25) becomes

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \hat{\mathbf{P}}_{in}^E \\ \hat{\mathbf{P}}_{in}^M \\ \lambda \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{F}}_n^E \\ \hat{\mathbf{F}}_n^M \\ \mathbf{0} \end{Bmatrix}. \tag{26}$$

Outer corner node. This case is as simple as the one with Dirichlet BC, but here instead of using the reaction force, the known external force $\mathbf{T}_n$ is applied. Once again the resulting force is obtained straightaway since the two forces are equilibrated. When no forces are applied, the node is treated as a free node and will only be affected by the corner node force $\hat{\mathbf{F}}_n^E$.

Reentrant corner node. Let us consider a reentrant corner formed by elements A, E and M in Fig. 5(d) with their known external forces $\mathbf{T}_n^A, \mathbf{T}_n^E, \mathbf{T}_n^M$. In this case, the Maxwell diagram becomes a triangle with the *pole* located at one vertex. Hence, the system of equations of equilibrium can be written as:

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \hat{\mathbf{P}}_{jn}^A \\ \hat{\mathbf{P}}_{in}^M \\ \hat{\mathbf{P}}_{in}^E \\ \hat{\mathbf{P}}_{jn}^E \\ \lambda \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{F}}_n^A - \mathbf{T}_n^A \\ \hat{\mathbf{F}}_n^M - \mathbf{T}_n^M \\ \hat{\mathbf{F}}_n^E - \mathbf{T}_n^E \\ \mathbf{0} \\ \mathbf{0} \end{Bmatrix}. \tag{27}$$

Thus, when no external forces are applied, Eq. (27) is reduced to

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \hat{\mathbf{P}}_{jn}^A \\ \hat{\mathbf{P}}_{in}^M \\ \hat{\mathbf{P}}_{in}^E \\ \hat{\mathbf{P}}_{jn}^E \\ \lambda \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{F}}_n^A \\ \hat{\mathbf{F}}_n^M \\ \hat{\mathbf{F}}_n^E \\ \mathbf{0} \\ \mathbf{0} \end{Bmatrix}. \tag{28}$$

3.3. Application of the SIMP method at the fine level

Once the equilibrated tractions at the coarse level have been obtained using the procedure previously described, we will solve another topology optimization problem in each element of this coarse level, considering these equilibrated tractions as Neumann BC and a fine-level discretization.

Since every fine-level problem is defined into a square domain, the SIMP method presented in [5] was used for the sake of simplicity, but slightly modified with the projection method proposed in [35], to solve the topology optimization problems at the fine level. A flowchart of the proposed procedure is shown in Fig. 7. Although the standard SIMP method is well-known, several comments are appropriate to clarify this flowchart:

a. As in the general procedure shown in Fig. 3, the density of each element e in the fine level at iteration k, is referred to as $\rho_e^{(k)}$.
b. In order to remove rigid body motions, simply support conditions have been adopted for the square domains that define the fine level. Since the lateral tractions applied on each cell are equilibrated, the support reactions vanish and do not affect to the internal behaviour at the fine level.
c. As the basic 99-line SIMP code introduced in [5] produced undesired grey areas in some areas of the cell, the exponential density projection (29) proposed by [35] with a density threshold of $\mu = 0.5$ was adopted to improve the convergence of the process:

$$\tilde{\rho}_e = \begin{cases} \mu[e^{-\beta(1-\rho_e/\mu)} - (1-\rho_e/\mu)e^{-\beta}] & 0 \le \rho_i \le \mu \\ (1-\mu)[1 - e^{-\beta(\rho_e-\mu)/(1-\mu)} \\ \quad +(\rho_e-\mu)e^{-\beta}/(1-\mu)] + \mu & \mu < \rho_i \le 1. \end{cases} \tag{29}$$

d. Focused on controlling the convergence more accurately, the projection of densities and the updating of the projection parameter $\beta^{(k)}$ are carried out every 2 iterations up to convergence. Indeed, $\beta^{(k)}$ is forced to remain below the value of the input parameter β_{max}. Otherwise, disjointed solutions with material regions separated by large void regions may appear.
e. A way to measure the level of sharpness obtained in the fine-level solution, the following parameter (see [36]) is calculated over each iteration:

$$M_{nd}^{(k)} = \sum_{e=1}^{n} \frac{4\rho_e^{(k)}(1-\rho_e^{(k)})}{n} 100\% \in [0, 100], \tag{30}$$

where n is the total number of elements in the fine-level FE mesh. As can be expected, very high or very low values of densities $\rho_e^{(k)}$ produce low values of $M_{nd}^{(k)}$, while intermediate values of $\rho_e^{(k)}$ produce high values of $M_{nd}^{(k)}$. Therefore, the parameter $M_{nd}^{(k)}$ is useful to identify fine-level solutions with too many elements having intermediate densities. Thus, if $M_{nd}^{(k)}$ is greater than certain value of the input parameter $M_{nd,min}$, the associated grey level is considered excessive and the corresponding density projection is carried out. Otherwise, the solution is considered to be sharp enough and the projection does not take place.
f. Although in the coarse level we considered $p = 1$ because we wanted to obtain a coarse representation of the density distribution, we use $p = 3$ in the fine-level topology optimization process to sharpen the final solution penalizing intermediate density values.

Table 1
Input parameters for example 1.

Coarse level	Fine level	Fine-level projection
32 × 16 elems.	32 × 32 elems.	$\beta_0 = 1$
$p = 1$	$p = 3$	$\beta_{max} = 2$
$r_{min} = 1.5$	$r_{min} = 1.3$	$\mu = 0.5$
$\varepsilon = 0.03$	$\varepsilon = 0.01$	$M_{nd,min} = 50\%$

4. Numerical results

The two-level topology optimization method is studied next considering two numerical examples under plane stress. The first one is based on the classic cantilever beam problem whereas the second one is a cantilevered L-shape domain problem that produces a singular stress field. Bilinear quadrilateral elements with Young Modulus E = 1000 MPa and poison ratio $\nu = 0.3$ were used in both problems.

4.1. Example 1. Cantilever beam under pure shear loading

In the first example the applied load at the free end of the beam is defined as a parabolic shear stress distribution with $\tau_{max} = 1$ MPa, as shown in Fig. 8. Beam dimensions are 2 × 1 m and the prescribed volume fraction is $\rho_0 = 0.5$. Other input parameters for this problem are outlined in Table 1. In the third column are those adopted in the fine-level Heaviside projection, whereas ε stands for the threshold of convergence for the coarse and fine-level TO procedures regarding the maximum change in the densities between two consecutive iterations.

Namely, the convergence criterion for the coarse topology optimization can be written:

$$\max \left| \rho_i^{(j+1)} - \rho_i^{(j)} \right| \le \varepsilon \tag{31}$$

where i corresponds to the chosen coarse element and (j) to the current iteration inside each coarse topology optimization process. Similarly, for the fine-level optimization, the convergence criterion is expressed:

$$\max \left| \rho_e^{(k+1)} - \rho_e^{(k)} \right| \le \varepsilon \tag{32}$$

where e is the chosen element and (k) is the current iteration.

A first test is considered to show the inter-cell continuity and the performance of the proposed method. The obtained smooth density distribution at the coarse level is shown in Fig. 9(a), where the number over the graph stands for the iterations up to convergence in the application of the SIMP method.

As expected, the material is accumulated on regions with high values of normal stresses due to the bending moment (darker elements on top and bottom regions), dark elements of high density are also obtained at the right-hand side, where the external load is applied. The central region of the domain appears light grey because of the low stress values produced by the shear force. Regarding the density distribution,

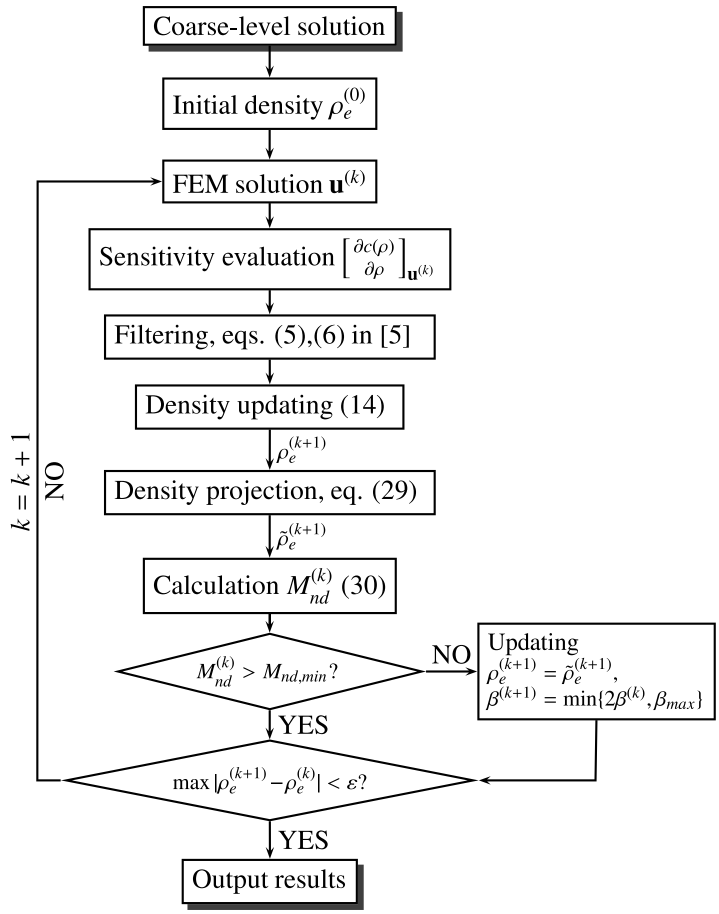


Fig. 7. Application of the SIMP method at the fine level.

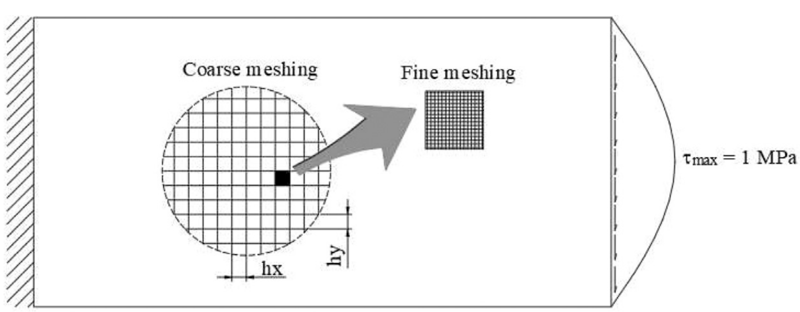


Fig. 8. Example 1, external loads and meshing.

note that the central left-hand region progressively tends to be a void region because of its low stress values.

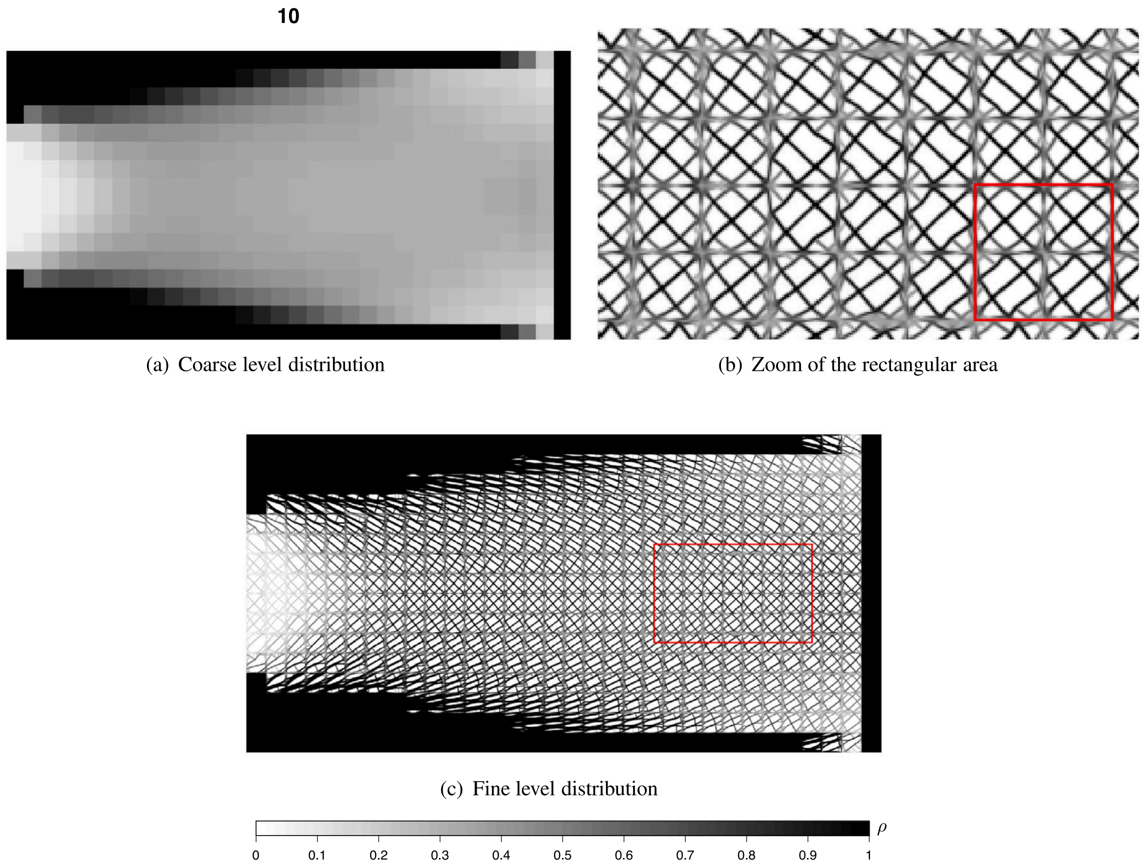
The trabecular configuration obtained after the fine-level topology optimization process is applied to each of the cells is shown in Fig. 9(c). The stress paths defined by the cross-like distribution of densities $\rho_e^{(k)}$ into each cell in the region around the neutral axis are similar to those shown in [37, Fig. 16(b)] for the same problem and to the results in [38, Fig. 3(b)] for a cantilever beam loaded with a point load at the free end. As can be seen, these paths change their slope becoming more horizontal as the distance to the neutral axis increases, i.e. as the ratio of the shear stress to the normal stress decreases.

Note that the applied load is skew-symmetric with respect the neutral axis, producing a material distribution mainly symmetrical. This is because the solution of the optimization problem seeks the minimum of the quadratic form $c(\rho)$, which is proportional to the strain energy density. Thus, there is no distinction between tension and compression in the final evaluation of stress.

As shown in Fig. 9(b), an acceptably-continuous inter-cell material distribution is obtained at the fine level. Despite this, since the lateral tractions act on the contour of each cell, the numerical behaviour of the fine-level problem produces a frame effect over the cell, leading to a grid-like distribution. The elimination of these artefacts will be object of future research.

To provide further insight into the qualitative influence of the fine-level mesh size on the continuity of the inner members across the surroundings of each cell boundary, three samples of the four cells marked in red in Fig. 9(b) are presented in Fig. 10 using three different fine-level meshing. As shown, finer meshes provide a reduction in the scope of this grid-like effect, along with a better qualitative continuity of the inner members across the cell boundary.

From the results of this first simulation, where some areas contain almost completely solid and almost completely void cells, and given the potentially high costs of manufacturing these kind of cells in relation

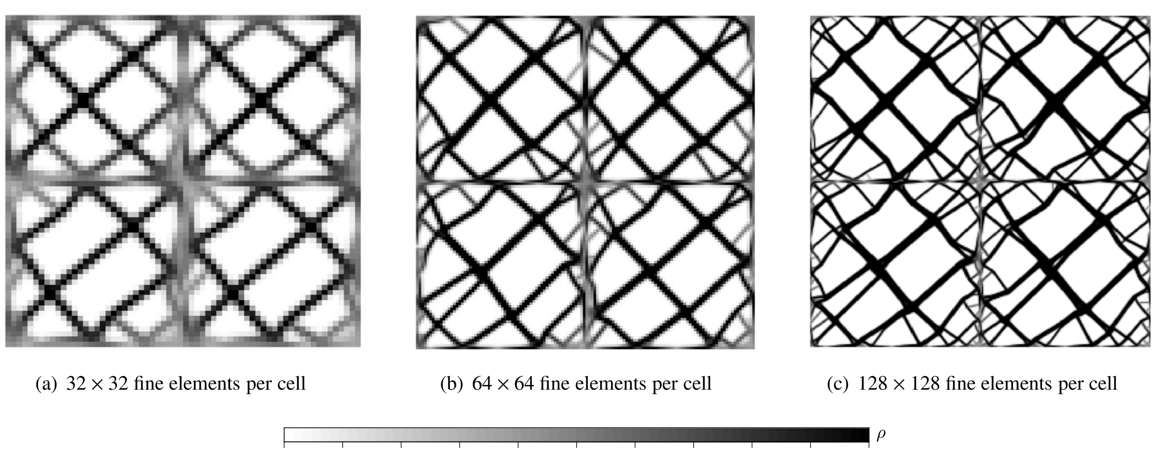


(a) Coarse level distribution

(b) Zoom of the rectangular area

(c) Fine level distribution

Fig. 9. Example 1, density distributions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



(a) 32×32 fine elements per cell

(b) 64×64 fine elements per cell

(c) 128×128 fine elements per cell

Fig. 10. Example 1, influence of the fine mesh size on the connectivity of inner members.

to completely solid (or void) cells, a new procedure able to avoid the former type of cells but maintaining the global volume fraction is desirable. Although using a Heaviside projection would be possible, as has been shown in [39] for rank-N laminates, some numerical issues could arise in relation with the parameter β_{max}. Namely, high values of this parameter are related to high slopes of the projection function for intermediate densities and provide solutions trending to $0-1$, but also might trigger numerical issues on the final convergence. On the other hand, the efficient removal-based scheme developed in [40] requires the evaluation of new boundary conditions when and element is removed. Consequently, a simple approach via fixing densities is proposed herein, even though certain computational efficiency could be sacrificed: the main idea is to convert almost solid and almost void elements into completely solid and completely void respectively.

Initially, a range of two prescribed density thresholds $[\overline{\rho}_{min}, \overline{\rho}_{max}]$ has been introduced in the general procedure shown in Fig. 3, providing three sets of elements: completely solid, completely void and intermediate-density (optimizable) elements. Thus, the updating procedure of densities (Fig. 11) works in the following way: after each optimization process, if the density of one element is outside the range $[\overline{\rho}_{min}, \overline{\rho}_{max}]$, it is added to a set of fixed elements. The elements in the list are shifted to 0 or 1, if they are beyond the respective threshold. The remaining elements with the remaining amount of material are

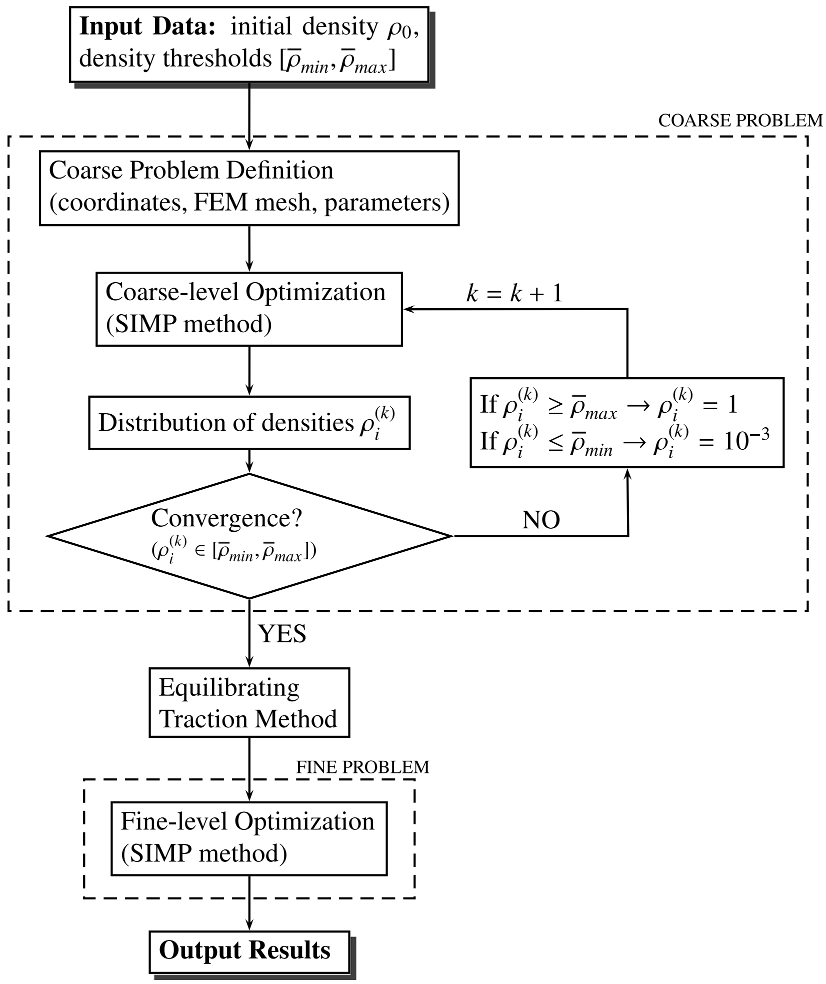


Fig. 11. Adaptation of the general procedure to additive manufacturing.

considered in a new optimization process. This process is repeated until no changes in the fixed element list are observed.

For the sake of completeness, further remarks about the equilibrium of internal nodes surrounding elements which have been turned into void, are described in Appendix.

This procedure allows to make two consecutive optimization problems numerically independent. This independency suggests that each round inside the iterative loop for the coarse problem (dashed line in Fig. 11) was referred to as "step" hereinafter, making a distinction from the term "iteration" used inside the own iterative process for each SIMP application.

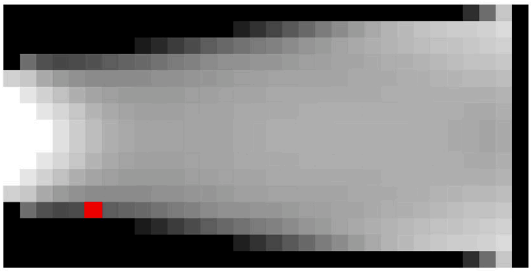
From the practical point of view, the possibility to set these thresholds allows the designer to adjust the method in order to provide designs with the desired amount of trabecular cells, that is the number of cells to be optimized in the fine level.

Aimed to verify the optimization performance of the proposed approach, a first analysis with $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.06, 0.94]$ has been carried out. The coarse and fine distributions of densities are represented in Figs. 12(a) and 12(b), while the objective functions in relative terms are represented in Figs. 12(d) and 12(e), for the coarse level and the fine level, respectively. Since each fine element has a significantly different convergence history, one element (in red in Fig. 12(a)) has been chosen as an example of the numerical performance at the fine level. In addition, the accumulated objective function for all the cells in the fine-level optimization, undergoes a relative reduction of 64.43% comparing its converged value with the fine-level initial one.

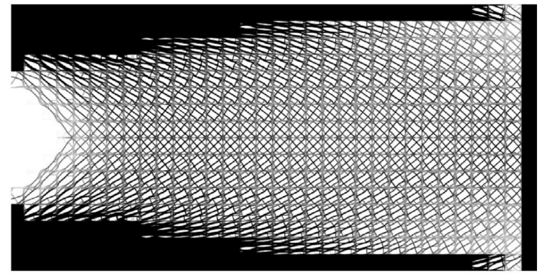
Accordingly, the computation time for the example in Fig. 12 was t = 3 h 35 min, run within our own MATLAB® code and using a computer with processor Intel® Core^{TM} $i9-10920X$ (3.5 GHz) and 256 Gb of RAM memory, without considering parallelization. Even if this time may give an order of magnitude of the computation time, it should be taken into account that it is deeply conditioned by other factors, like the programmer skills or the choice of the programming language.

For the purpose of qualitatively studying the influence of density thresholds $[\overline{\rho}_{min}, \overline{\rho}_{max}]$ on the final distributions of material, three coarse-level converged solutions with different relative density ranges have been represented in Figs. 13(a), 13(c), 13(e) together with the corresponding fine-level solutions in Figs. 13(b), 13(d), 13(f). As shown, the solution becomes closer to the classical optimization problem as the density range narrows (see [10,38,41] as examples). Accordingly, Figs. 13(e) and 13(f) represent an almost solid-and-void solution for the narrowest interval, where only a few number of cells are optimized at the fine level. These results allow to validate the proposed approach for the coarse level.

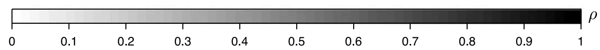
In addition, the effect of using non-symmetric density ranges is shown in Fig. 14 by comparing results from a symmetric gap (Figs. 14(c) and 14(d)) with other from non-symmetric ones. Comparing Figs. 14(a) and 14(c), if only the upper density threshold $\overline{\rho}_{max}$ is reduced, the size of the solid region is increased, although not remarkably. However, comparing 14(c) and 14(e), if only the lower density threshold $\overline{\rho}_{min}$ is increased to the same extent, the size of the void regions is increased significantly. Therefore, we can observe that the effect of reducing $\overline{\rho}_{max}$ is qualitatively smaller than that of increasing $\overline{\rho}_{min}$ because there are large areas with a relatively low value of density in this problem. Despite this, the effects of modifying either $\overline{\rho}_{min}$ or $\overline{\rho}_{max}$ are not completely uncoupled, since the total amount of material into the domain should

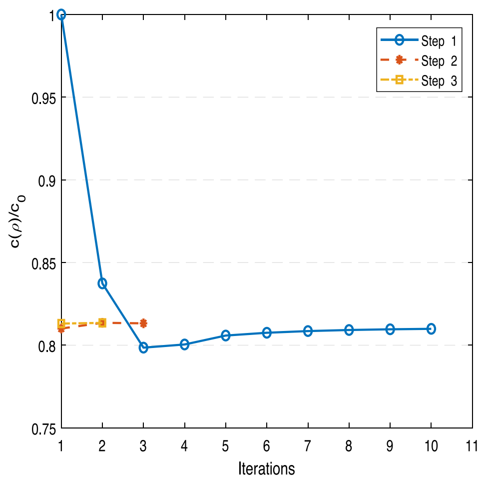


(a) Coarse density distribution. Location of element 122

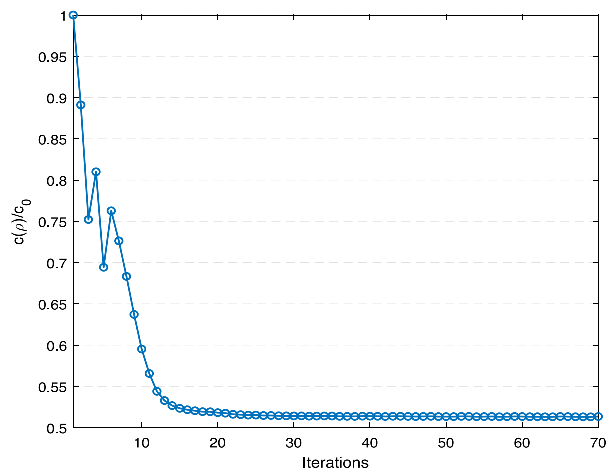


(b) Final density distribution





(d) Objective function at coarse level



(e) Objective function element 122

Fig. 12. Example 1, final results for $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.06, 0.94]$. The three curves in 12(d) indicate the three optimization processes required to enforce the element outside the range to 0 or 1. The blue line is the first optimization process, requiring more iterations, while the other two are two subsequent optimization processes. Since only small variations appear compared to the previous step, the convergence is fast. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

be kept equal to $\overline{\rho}_0$ and the whole density distribution is affected by modifying one of the two ends of the range.

To conclude this first example, the influence of the coarse element size on the final results is studied. On the basis of a density range of $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.12, 0.88]$, three coarse mesh sizes are compared by using 16 × 8, 32 × 16 and 64 × 32 coarse meshes respectively for the same design domain. For each simulation, the converged coarse solution and the final solution are shown in Fig. 15. In order to keep the final discretization level, the fine mesh has been properly tuned in such a way the number of total fine elements are the same in each direction for the three performed simulations.

As can be seen, the results show good robustness of the method against variations of the coarse mesh size. Final solutions are qualitatively similar and stable for different meshing. However, as can be expected, the final trabecular topology is highly dependent on the mesh size. Even void and solid areas may experience different distributions under mesh size variations, as can be checked in the right upper and bottom corners of Figs. 15(e) and 15(f).

4.2. Example 2. Cantilevered L-shape domain under bending load

Our second example is a cantilevered L-shape domain under a parabolic vertical load, as shown in Fig. 16, that produces a stress singularity at point Q. Outer dimensions of the domain are 10 × 10 m. The prescribed volume fraction is $\rho_0 = 0.5$. Other input parameters are outlined in Table 2 (see definitions (31) and (32)).

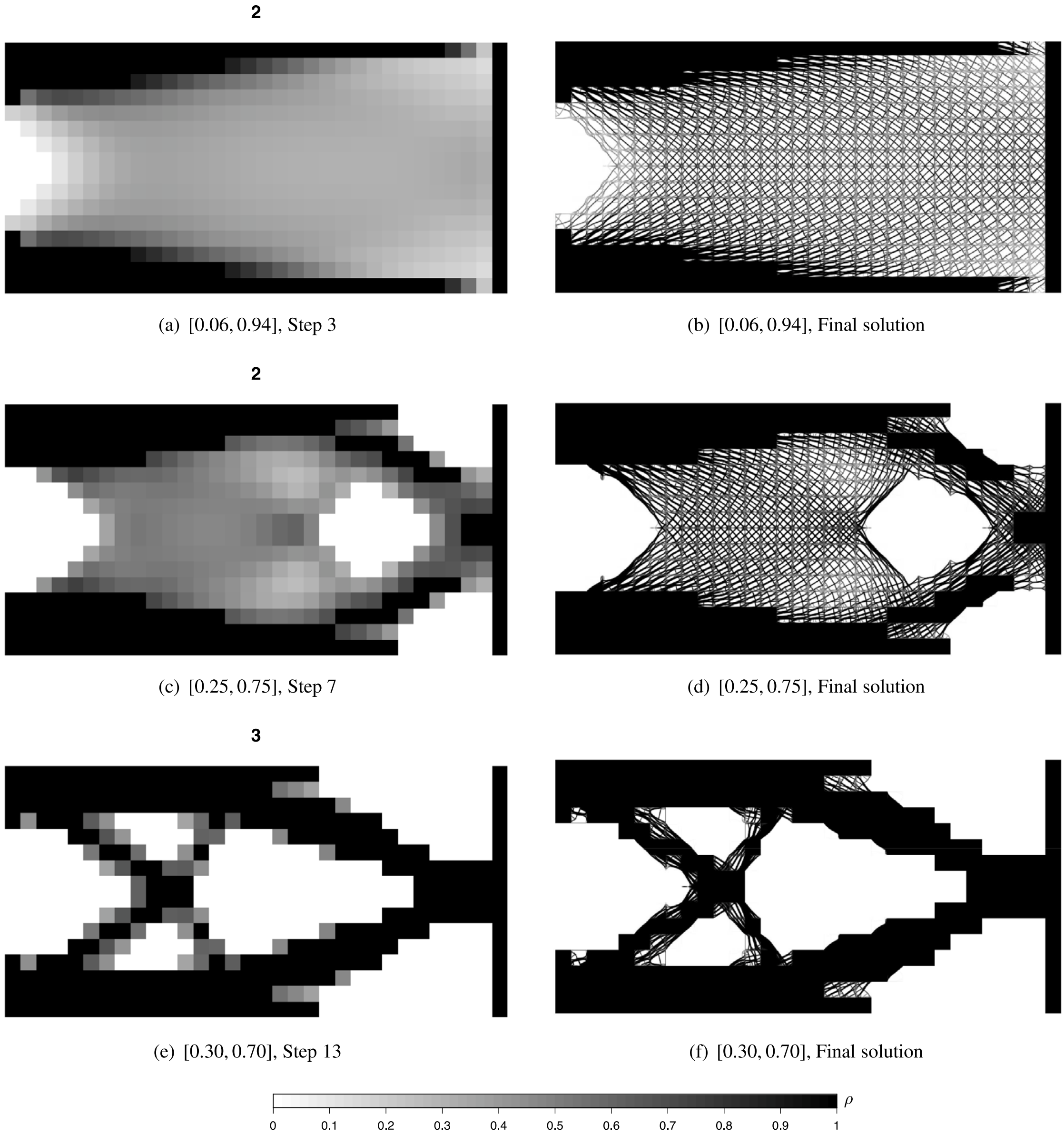
Table 2
Input parameters for example 2.

Coarse level	Fine level	Fine-level projection
32 × 32 elems.	32 × 32 elems.	$\beta_0 = 1$
$p = 1$	$p = 3$	$\beta_{max} = 4.5$
$r_{min} = 1.5$	$r_{min} = 1.3$	$\mu = 0.5$
$\varepsilon = 0.03$	$\varepsilon = 0.01$	$M_{nd,min} = 45\%$

A first test with $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.12, 0.88]$ is considered. Several steps of the coarse-level iterative procedure are plotted in Fig. 17, where Fig. 17(c) represents the converged solution.

Due to the bending effect of the external load over the L-shaped domain, significant accumulations of material are noticed on the outer sides of the left upper region of the domain. In addition, remarkable accumulations can be observed around the singularity at point Q and near the boundary where the external load is applied. Even into the light-grey area, a slightly darker arch can be distinguished indicating the curved load path from the external load to the clamped end. Otherwise, void regions appear in the central zone of the clamped end and around the bottom corners of the domain, due to the reduced stress levels on these regions.

Similarly to the first example, the optimization performance of the method is shown via the relative objective functions at the coarse and fine level for a density range of $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.12, 0.88]$ in Fig. 19. The convergence history of element 140 (in red in Fig. 19(a)) is plotted as an example in Fig. 19(c). Accordingly, the accumulated objective



(a) [0.06, 0.94], Step 3
(b) [0.06, 0.94], Final solution
(c) [0.25, 0.75], Step 7
(d) [0.25, 0.75], Final solution
(e) [0.30, 0.70], Step 13
(f) [0.30, 0.70], Final solution

Fig. 13. Example 1, effect of different density thresholds on the relative density distribution.

function for all the cells in the fine-level optimization, experiences a relative reduction of 54.89% comparing its converged value with the fine-level initial one.

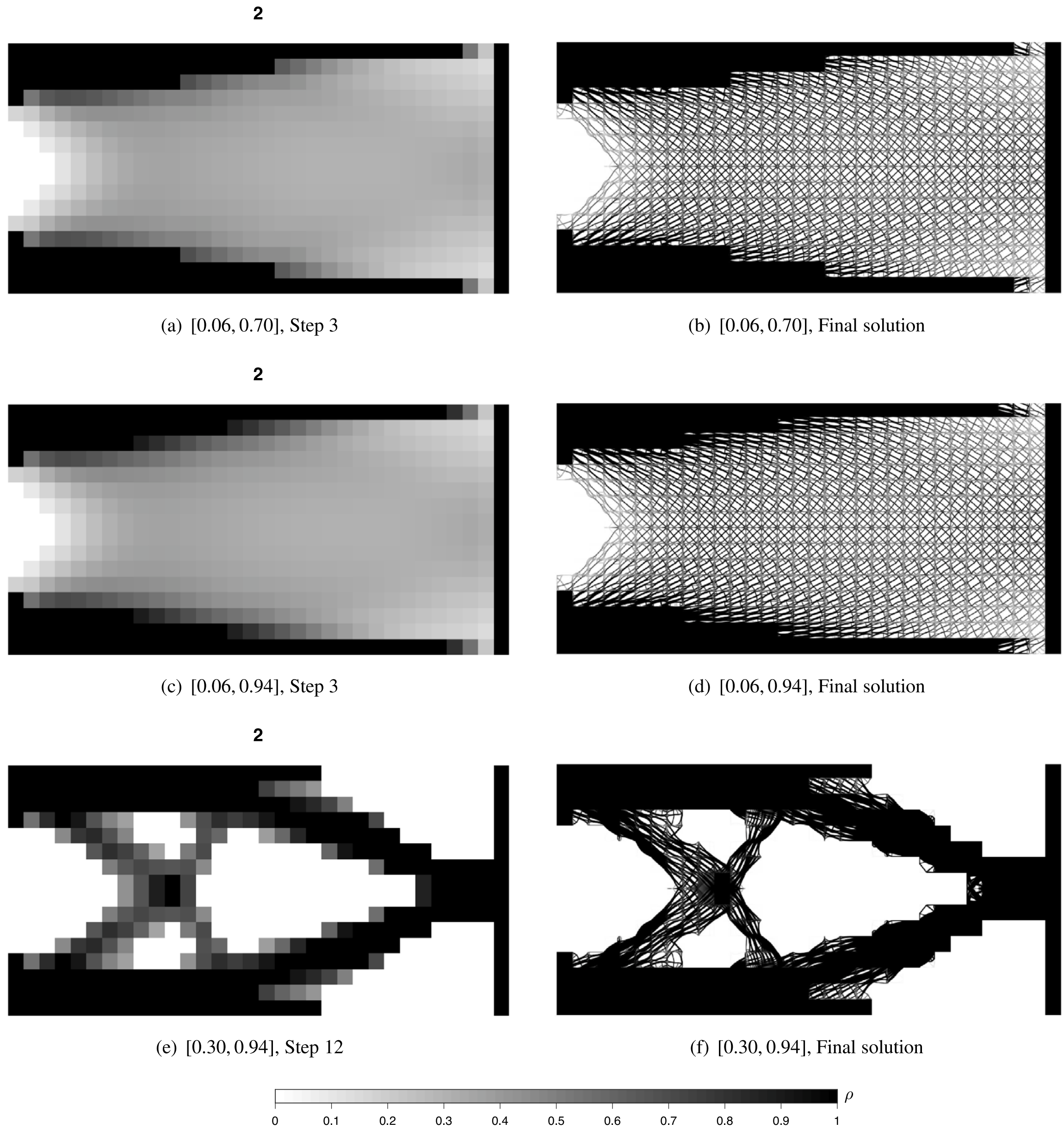
After the coarse problem has been solved, the solution of the fine-level problem is carried out (see Fig. 18). As shown, paths of principal stresses in the lower region of the L-shape domain define arches and radii centred at point Q, as in [37, Fig. 17], [42, Fig. 10], [43, Fig. 8] and [44, Tabs. 3,4]. In the vertical branch of the beam these paths are nearly vertical because of the higher significance of normal stresses with respect to shear stresses.

As for example 1, the qualitative influence of density thresholds on the final solution is studied considering three different ranges of density thresholds (see Fig. 20). For each simulation, the converged coarse-level (Figs. 20(a), 20(c), 20(e)) and their respective fine-level solutions (Figs. 20(b), 20(d), 20(f)) are displayed. As in example 1, the smaller the range of density thresholds, the larger the total size of solid and void regions. Therefore, for very small intervals, the fine-level solution will tend to the classical topology optimization solution (see [43,44] as examples).

Additionally, simulations with non-symmetric gaps (Figs. 21(a), 21(c), 21(d), 21(f)) have been compared to results with a symmetric gap in Figs. 21(c), 21(e), obtaining similar trends to those obtained for example 1, i.e., when $\overline{\rho}_{max}$ is reduced, the size of solid regions is incremented and when $\overline{\rho}_{min}$ is increased, void regions become larger too.

5. Concluding remarks

In the present work, a two-level topology optimization technique based on the SIMP method is proposed. Firstly, a topology optimization problem is solved at the coarse level by using the FEM, which determines the general distribution of material in the component. As a result, a constant relative density value is assigned to each coarse-level cell represented by one finite element. Secondly, after evaluating an equilibrated traction field by means of the traction equilibrating method, a new topology optimization problem is solved into each cell at the fine level. Finally, after compiling the solution of each individual cell into one single result, a reasonably continuous across-cells material distribution is obtained. In order to show the robustness of the method,



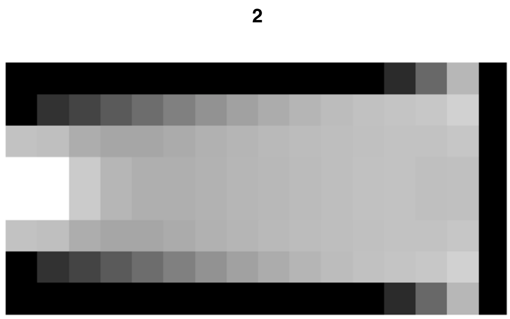
(a) [0.06, 0.70], Step 3
(b) [0.06, 0.70], Final solution
(c) [0.06, 0.94], Step 3
(d) [0.06, 0.94], Final solution
(e) [0.30, 0.94], Step 12
(f) [0.30, 0.94], Final solution

Fig. 14. Example 1, effect of non-symmetric density ranges on the relative density distribution.

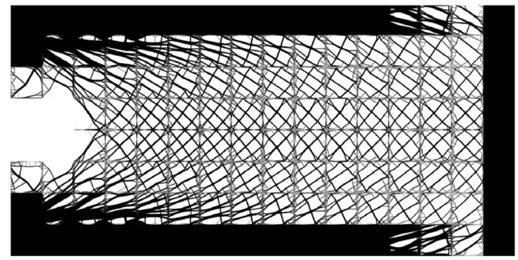
two numerical examples have been presented: a cantilever beam under shear loading and a L-shape domain under bending load. Both have shown good agreement with other published works [10,37,38,41–44] and significant sharpness level.

The proposed methodology provides high-resolution trabecular solutions of the optimal distribution of material into the design domain with geometrical continuity through a two-level topology optimization method, where the continuity of tractions across the coarse-level elements has been imposed. The combination of two existing methodologies, the traction equilibrating method [27] and the SIMP method [3,5], to uncouple the topology optimization of each individual cell represents the key aspect in the development of this research. In fact, this numerical uncoupling allows to significantly reduce the required computing time as has been explained in Section 3.1, even it can speed up the process by means of numerical parallelization techniques. This features makes the proposed two-level approach readily scaleable to larger domains at a reasonable numerical expense.

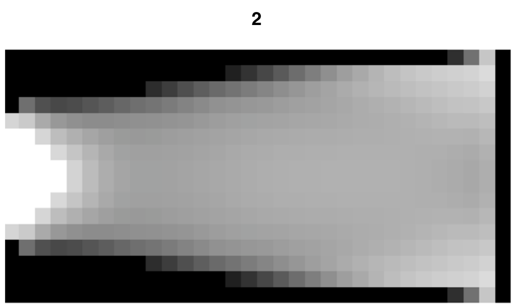
The simplicity of the presented methodology has been properly adapted to the features of additive manufacturing. Namely, a range of prescribed density thresholds has been introduced to transform almost void ($\rho_i^{(k)} < \overline{\rho}_{min}$) or almost solid ($\rho_i^{(k)} > \overline{\rho}_{max}$) cells into completely void or completely solid cells in the final solution. This new methodology not only reduces the total number of cells on which to perform fine-level optimization, but also reduces the manufacturing cost by avoiding the manufacturing of nearly void or nearly solid cells. In fact, making certain elements completely void or solid does not substantively constrain the whole design of the component. The numerical results show that the narrower the range of density thresholds, the closer is the final result to the classic SIMP solutions, with a greater relative importance of the coarse level compared to the fine level in the final solution. In fact, narrow density ranges can produce noticeable deviations from the two-level optimum solution obtained with wide ranges. On the other hand, regarding the non-symmetry of the range of density thresholds,



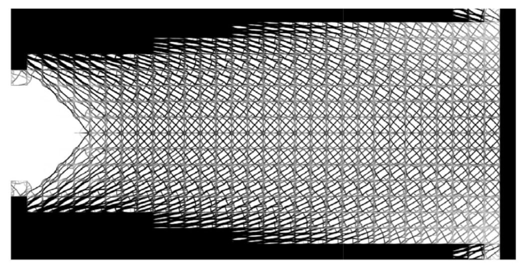
(a) 16 × 8 coarse elems., Step 2



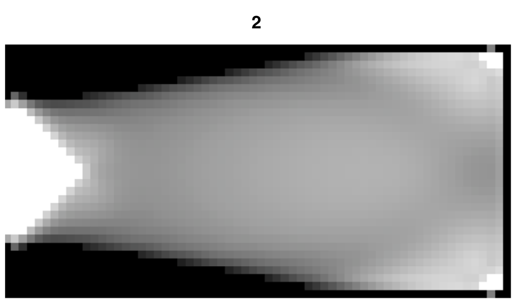
(b) 16 × 8 coarse elems., Final solution



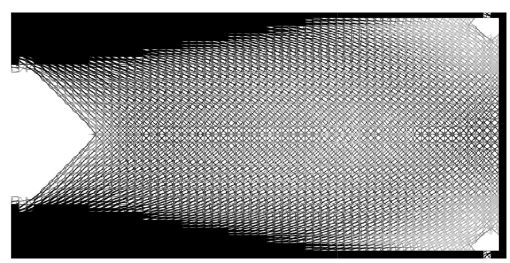
(c) 32 × 16 coarse elems., Step 3



(d) 32 × 16 coarse elems., Final solution



(e) 64 × 32 coarse elems., Step 8



(f) 64 × 32 coarse elems., Final solution

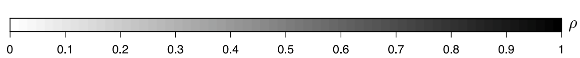


Fig. 15. Example 1, effect of the coarse element size with $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.12, 0.88]$.

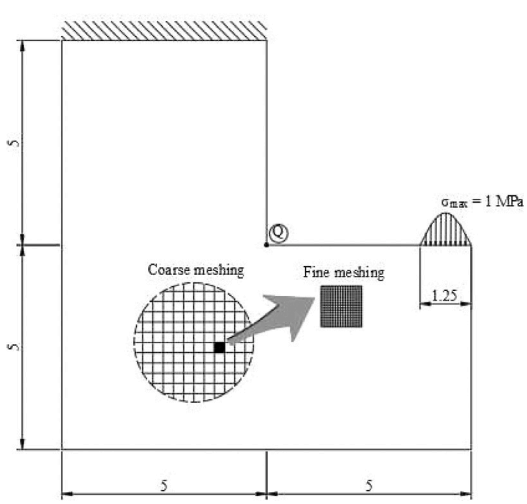


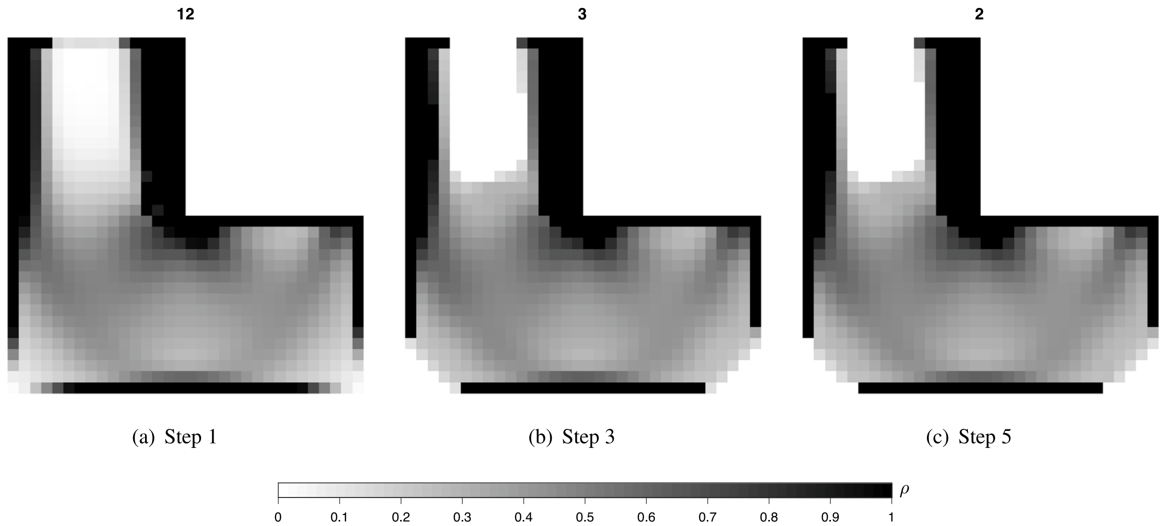
Fig. 16. Example 2, external loads and meshing.

the presented solutions are qualitatively more sensitive to an increase of $\overline{\rho}_{min}$ (larger void regions) than to a decrease of $\overline{\rho}_{max}$ (larger solid regions).

Several limitations are object of future research, such as including the homogenization theory in the coarse level to evaluate the equivalent mechanical properties of each cell in the coarse optimization problem. So far, neglecting in the coarse problem that anisotropy of the fine level means that just near-optimal solutions can be obtained. Accordingly, the obtained topologies are highly dependent on the coarse mesh size with grid-like accumulation of material over the boundary of each cell, which has to be further analysed. Moreover, the assumption of constant density into each coarse cell is a source of discontinuity, but since the density variation in the coarse level is smooth, this fact has not a great influence.

The presented multilevel method can be taken as a starting point to improve the mechanical connectivity of multi-scale methods [25]. In fact, once developed for 3D applications, the proposed methodology has the potential to become a design tool for industrial environments, especially for additive manufacturing applications [45].

To sum up, the proposed methodology is a step forward to obtain trabecular-like structures at a reasonable computational cost that could



(a) Step 1 (b) Step 3 (c) Step 5

Fig. 17. Example 2, density distribution of the coarse problem with $[\overline{\rho}_{min}, \overline{\rho}_{max}][0.12, 0.88]$.

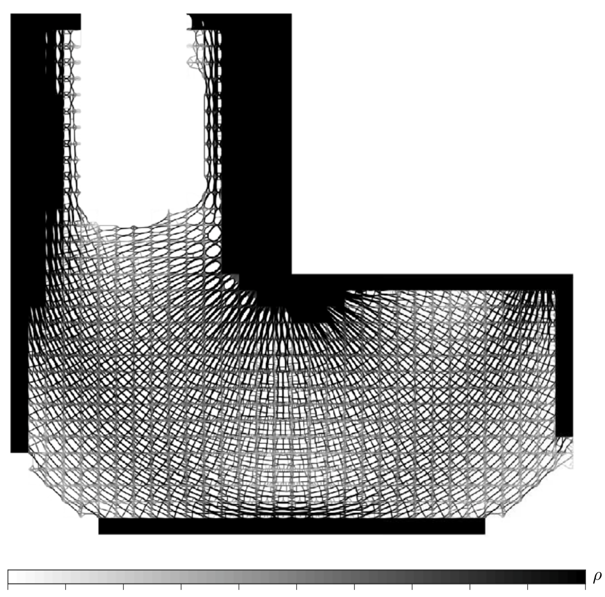


Fig. 18. Example 2, high-definition relative density distribution with $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.12, 0.88]$.

be used in traditional applications to generate structurally sound infills with additive manufacturing techniques.

CRediT authorship contribution statement

Rafael Merli: Writing – original draft, Software, Methodology, Investigation. **Antolín Martínez-Martínez:** Writing – review & editing, Methodology, Investigation. **Juan José Ródenas:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Marc Bosch-Galera:** Writing – review & editing, Methodology, Investigation. **Enrique Nadal:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

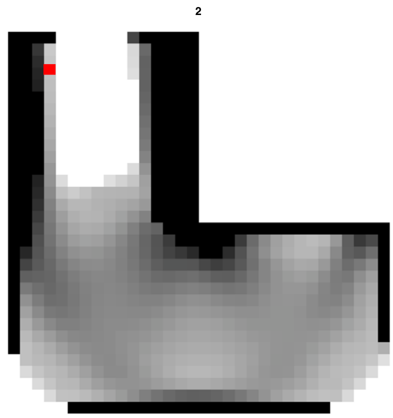
The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rafael Merli reports administrative support was provided by Spain Ministry of Science and Innovation. Antolin Martinez-Martinez reports financial support was provided by Government of Spain Ministry of Education and Vocational Training. Marc Bosch-Galera reports financial support was provided by Government of Spain Ministry of Education and Vocational Training. Juan Jose Rodenas reports financial support was provided by Spain Ministry of Science and Innovation. Enrique Nadal reports financial support was provided by Generalitat Valenciana.

Acknowledgements

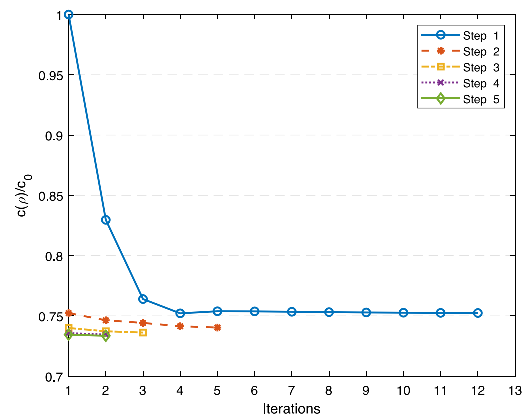
The authors gratefully acknowledge the support provided by Generalitat Valenciana, Spain (Prometeo/2021/046), Vicerrectorado de Investigación de la Universitat Politècnica de València (PAID-11-22), Ministerio de Ciencia e Innovación, Spain, Agencia Estatal de Investigación and FEDER, Spain (PID2022-141512NB-I00), as well as by Ministerio de Educación y Formación Profesional, Spain (FPU19/02103 and FPU20/04828). Furthermore, the authors wish to thank Mikel Barral for his seminal contribution to this work.

Appendix. Special cases of equilibrium

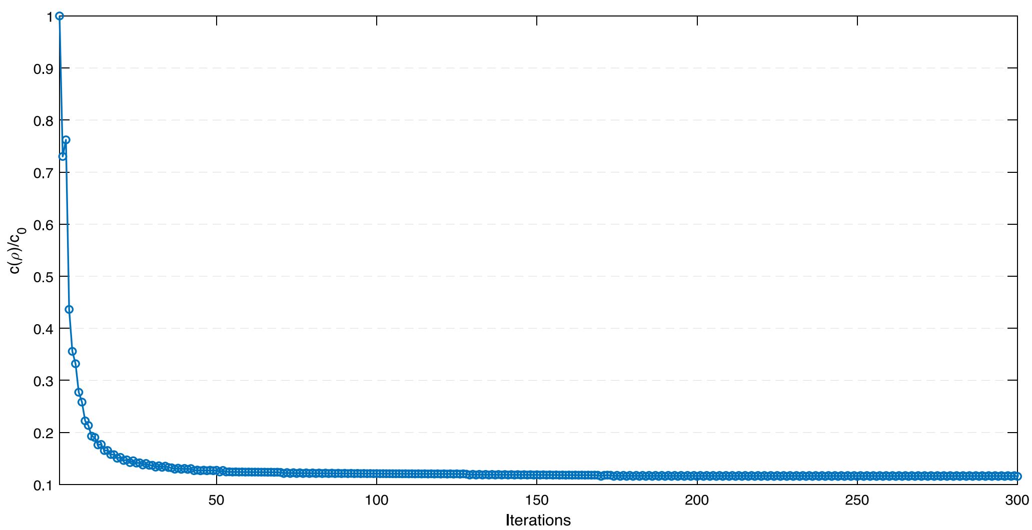
The internal nodes neighbouring elements which have been turned into void (assigning them $\rho = 10^{-3}$) in the coarse problem iterative procedure included in the algorithm described in Fig. 3, are considered special cases of equilibrium. In fact, if one or more elements around a node become void, the edge between void and non-void elements becomes equivalent to a Neumann boundary subjected to null tractions (Fig. A.22). However, if the general forces polygon shown in Fig. 6(a) is used and the *pole* is kept at the geometrical mass centre of the polygon, it is possible than a vanishing nodal force could produce corresponding non-vanishing side forces, yielding undesired accumulations of material



(a) Location of element 140



(b) Objective function at coarse level



(c) Objective function element 140

Fig. 19. Example 2, convergence history for $[\overline{\rho}_{min}, \overline{\rho}_{max}] = [0.12, 0.88]$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

along the new Neumann-like boundary in the fine-level optimization. In order to overcome this issue, further analysis is appropriate.

Let us consider element E to be optimized, included in the shaded area of Fig. A.23. It should be noted that only one or two void edge neighbours can be considered. Otherwise, if three edges of an ordinary element are neighbouring void elements, the global equilibrium of moments would not be feasible. In that case, element E would be turned into void (even if its density is into the density range) in order to keep the whole equilibrium of the component.

Some examples of special cases involving void elements and the order of elements adopted in our implemented code to solve the equilibrium of nodes are represented in Fig. A.22. The particular features of each case are explained below.

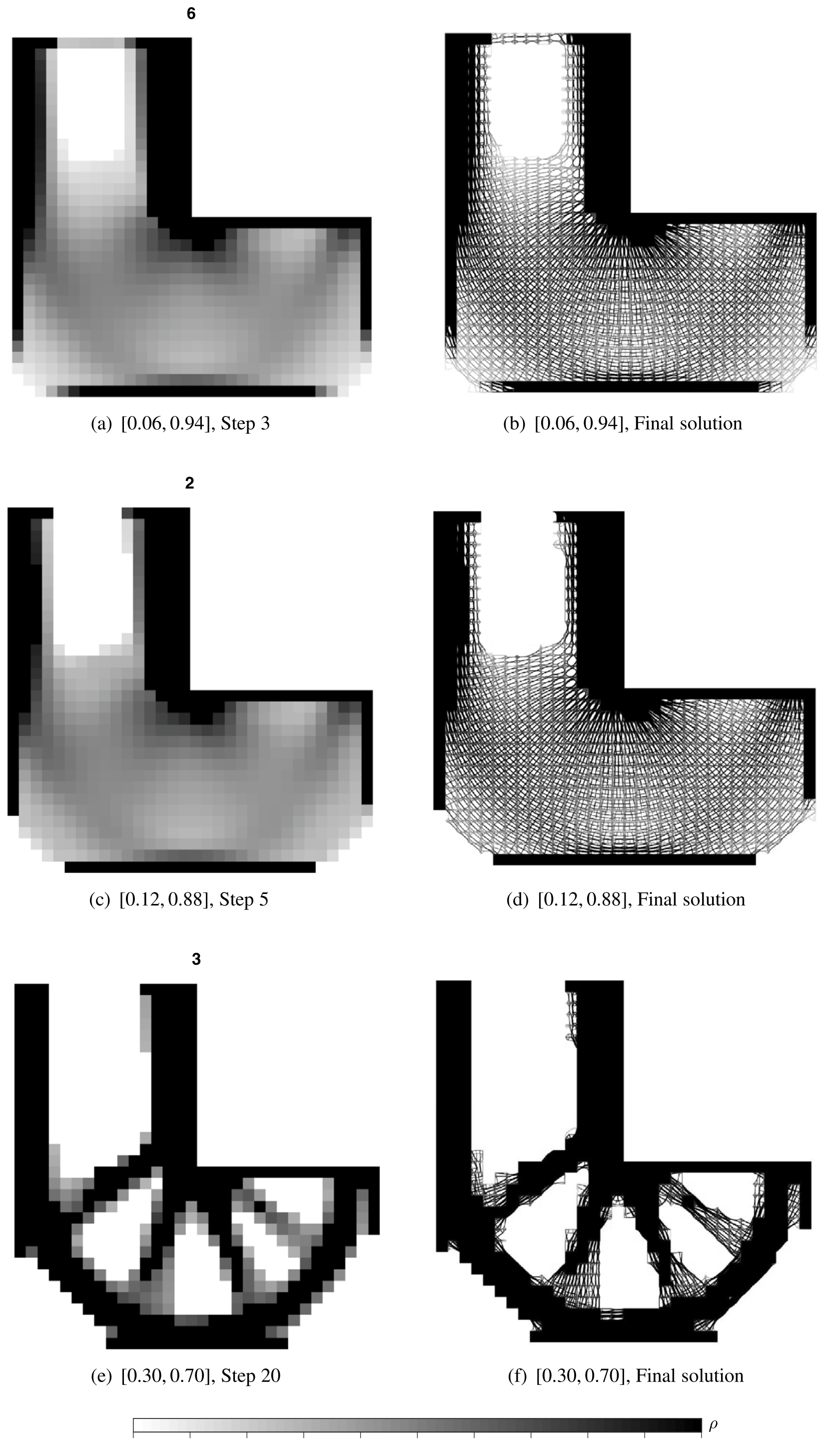
One-void-neighbour node. In this case, only a nodal force around the node will be significantly smaller than the rest of them. Therefore, the quadrilateral becomes similar to a triangle (see Fig. A.24 as an example). The pole will be located at the midpoint of the vanishing nodal force in order to obtain negligible side forces associated to the node of the void element.

Two-void-neighbour node. Nodes with two adjacent void elements around the node are included into this group. Thus, two consecutive (excluding the singular checkerboard patterns) nodal forces of the polygon are smaller than the other by several orders of magnitude and the quadrilateral becomes a needle-like polygon (see Fig. A.24). Extending the idea exposed in the previous case, the *pole* is located at the vertex between the two neglecting nodal forces.

Three-void-neighbour node. In the latter special case, nodes with three void elements around the node are considered. The polygon in this case is a quadrilateral of arbitrary shape. Here, all the forces are of small magnitude and similar to each other. In order to reduce the side forces of the non-void element (B in the case represented in Fig. A.24), the *pole* is located at the midpoint of its corresponding nodal force.

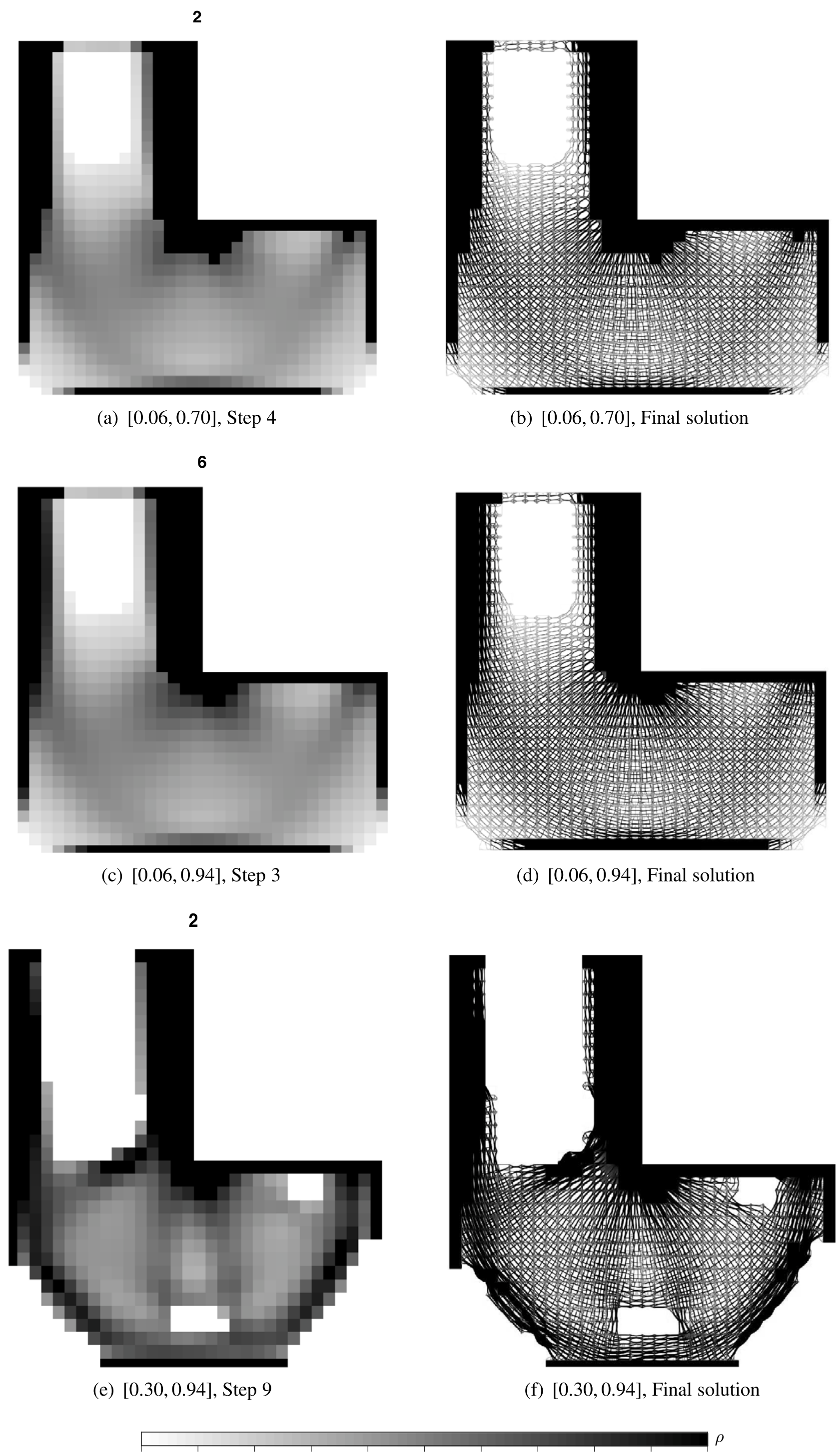
Data availability

No data was used for the research described in the article.



(a) [0.06, 0.94], Step 3
(b) [0.06, 0.94], Final solution
(c) [0.12, 0.88], Step 5
(d) [0.12, 0.88], Final solution
(e) [0.30, 0.70], Step 20
(f) [0.30, 0.70], Final solution

Fig. 20. Example 2, effect of different density thresholds on the relative density distribution.



(a) [0.06, 0.70], Step 4

(b) [0.06, 0.70], Final solution

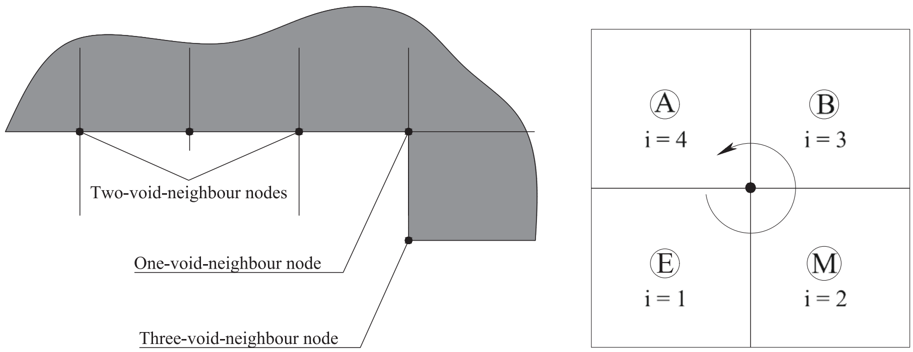
(c) [0.06, 0.94], Step 3

(d) [0.06, 0.94], Final solution

(e) [0.30, 0.94], Step 9

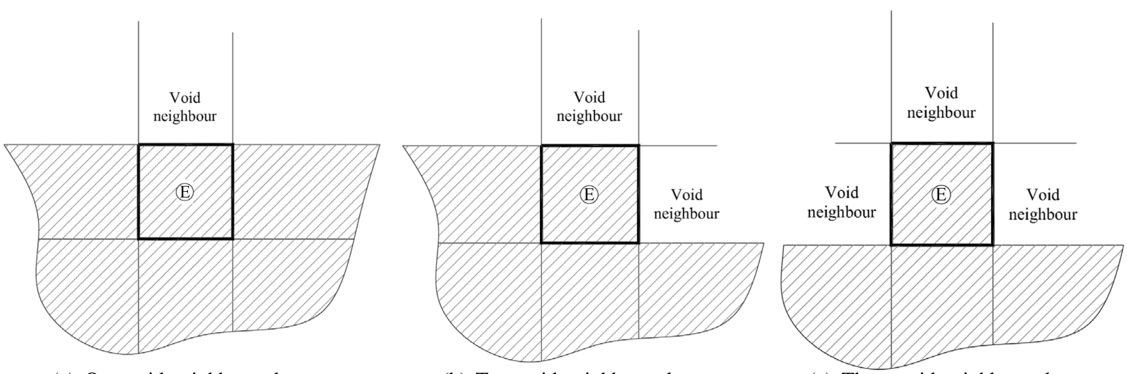
(f) [0.30, 0.94], Final solution

Fig. 21. Example 2, effect of non-symmetric density ranges on the relative density distribution.



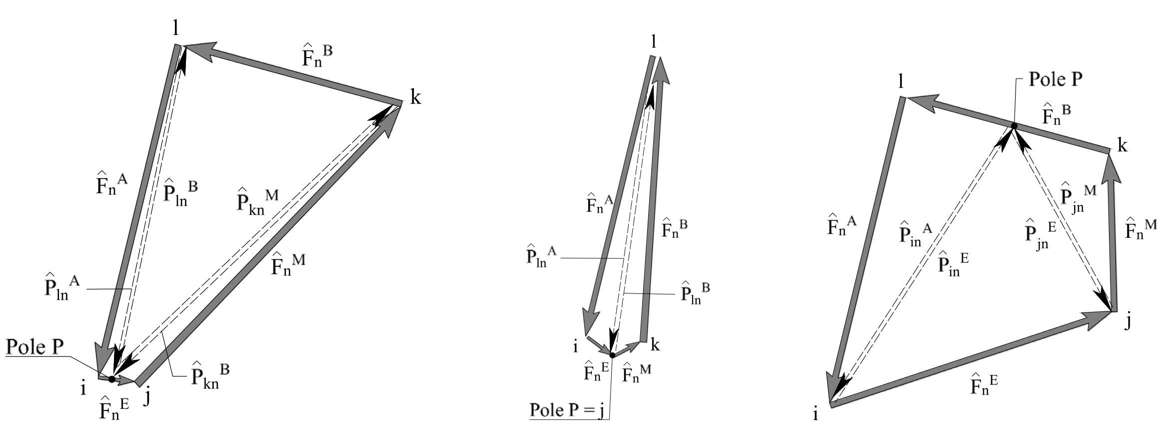
(a) Sample of different cases (b) Ordering of elements around the node

Fig. A.22. Special cases of nodes neighbouring void elements.



(a) One-void-neighbour element (b) Two-void-neighbour element (c) Three-void-neighbour element

Fig. A.23. Cases of neighbouring through edges.



(a) One-void-neighbour node (b) Two-void-neighbour node (c) Three-void-neighbour node

Fig. A.24. Special cases for Maxwell polygon.

References

[1] Marco O, Ródenas JJ, Albelda J, Nadal E, Tur M. Structural shape optimization using cartesian grids and automatic h-adaptive mesh projection. Struct Multidiscip Optim 2018;58:61–81, URL https://doi.org/10.1007/s00158-017-1875-1.

[2] Bendsøe MP, Kikuchi N. Generating optimal topologies in structural design using a homogenization method. Comput Methods Appl Mech Engrg 1988;71:197–224, URL https://doi.org/10.1016/0045-7825(88)90086-2.

[3] Bendsoe MP, Sigmund O. Material interpolation schemes in topology optimization. Arch Appl Mech 1999;69:635–54, URL https://doi.org/10.1007/10.1007/s004190050248.

[4] Bendsoe M, Sigmund O. Topology optimization: theory, methods, and applications. 2nd ed.. Springer Berlin Heidelberg; 2003.

[5] Sigmund O. A 99 line topology optimization code written in matlab. Struct Multidiscip Optim 2001;21:120–7, URL https://doi.org/10.1007/s001580050176.

[6] Andreassen E, Clausen A, Schevenels M, Lazarov BS, Sigmund O. Efficient topology optimization in MATLAB using 88 lines of code. Struct Multidiscip Optim 2011;43:1–16. http://dx.doi.org/10.1007/s00158-010-0594-7.

[7] Ferrari F, Sigmund O. A new generation 99 line matlab code for compliance topology optimization and its extension to 3D. Struct Multidiscip Optim 2020;62:2211–28, URL https://doi.org/10.1007/s00158-020-02629-w.

[8] Xie YM, Steven GP. A simple evolutionary procedure for structural optimization. Comput Struct 1993;49:885–96, URL https://doi.org/10.1016/0045-7949(93)90035-C.

[9] Wang MY, Wang X, Guo D. A level set method for structural topology optimization. Comput Methods Appl Mech Engrg 2003;192:227–46, URL https://doi.org/10.1016/S0045-7825(02)00559-5.

[10] Allaire G, Gournay FD, Jouve F, Toader AM. Structural optimization using topological and shape sensitivity via a level set method. Control Cybern 2005;34:59–80, URL http://eudml.org/doc/209353.
[11] Eschenauer HA, Olhoff N. Topology optimization of continuum structures: A review. Appl Mech Rev 2001;54:331–90, URL https://doi.org/10.1115/1.1388075.
[12] Wu J, Sigmund O, Groen JP. Topology optimization of multi-scale structures: a review. Struct Multidiscip Optim 2021;63:1455–80, URL https://doi.org/10.1007/s00158-021-02881-8.
[13] Smit T, Koppen S, Ferguson SJ, Helgason B. Conceptual design of compliant bone scaffolds by full-scale topology optimization. J Mech Behav Biomed Mater 2023;143:105886, URL https://doi.org/10.1016/j.jmbbm.2023.105886.
[14] Nguyen TH, Paulino GH, Song J, Le CH. A computational paradigm for multiresolution topology optimization. Struct Multidiscip Optim 2010;41(4):525–39, URL https://doi.org/10.1007/s00158-009-0443-8.
[15] Coelho PG, Fernandes PR, Guedes JM, Rodrigues HC. A hierarchical model for concurrent material and topology optimisation of three-dimensional structures. Struct Multidiscip Optim 2008;35:107–15, URL https://doi.org/10.1007/s00158-007-0141-3.
[16] Avellaneda M. Optimal bounds and microgeometries for elastic two-phase composites. SIAM J Appl Math 1987;47:1216–28, URL https://doi.org/10.1137/0147082.
[17] Bendsøe MP. Optimal shape design as a material distribution problem. Struct Optim 1989;1:193–202, URL https://doi.org/10.1007/BF01650949.
[18] Allaire ZBG, Jouve F. The homogenization method for topology and shape optimization. Single and multiple loads case. Rev Eur Finish 1996;5:649–72, URL https://doi.org/10.1080/12506559.1996.10511241.
[19] Träff E, Sigmund O, Groen JP. Simple single-scale microstructures based on optimal rank-3 laminates. Struct Multidiscip Optim 2019;59:1021–31, URL https://doi.org/10.1007/s00158-018-2180-3.
[20] Wu Z, Xia L, Wang S, Shi T. Topology optimization of hierarchical lattice structures with substructuring. Comput Methods Appl Mech Engrg 2019;345:602–17, URL https://doi.org/10.1016/J.CMA.2018.11.003.
[21] Wang C, Zhu JH, Zhang WH, Li SY, Kong J. Concurrent topology optimization design of structures and non-uniform parameterized lattice microstructures. Struct Multidiscip Optim 2018;58:35–50, URL https://doi.org/10.1007/s00158-018-2009-0.
[22] Wang F, Sigmund O. Numerical investigation of stiffness and buckling response of simple and optimized infill structures. Struct Multidiscip Optim 2020;61:2629–39, URL https://doi.org/10.1007/s00158-020-02525-3.
[23] Zhu Y, Li S, Du Z, Liu C, Guo X, Zhang W. A novel asymptotic-analysis-based homogenisation approach towards fast design of infill graded microstructures. J Mech Phys Solids 2019;124:612–33, URL https://doi.org/10.1016/j.jmps.2018.11.008.
[24] Ferrer A, Oliver J, Cante JC, Lloberas-Valls O. Vademecum-based approach to multi-scale topological material design. Adv Model Simul Eng Sci 2016;3:1–22, URL https://doi.org/10.1186/s40323-016-0078-4.
[25] Garner E, Kolken HMA, Wang CCL, Zadpoor AA, Wu J. Compatibility in microstructural optimization for additive manufacturing. Addit Manuf 2019;26:65–75, URL https://doi.org/10.1016/j.addma.2018.12.007.
[26] Liu P, Kang Z, Luo Y. Two-scale concurrent topology optimization of lattice structures with connectable microstructures. Addit Manuf 2020;36:101427, URL https://doi.org/10.1016/j.addma.2020.101427.
[27] Ladevèze P, Maunder EA. A general method for recovering equilibrating element tractions. Comput Methods Appl Mech Engrg 1996;137:111–51, URL https://doi.org/10.1016/S0045-7825(96)01067-5.
[28] Muñoz D, Albelda J, Ródenas JJ, Nadal E. Improvement in 3D topology optimization with h-adaptive refinement using the cartesian grid finite element method. Internat J Numer Methods Engrg 2022;123:3045–72, URL https://doi.org/10.1002/nme.6652.
[29] Krog LA, Olhoff N. Optimum topology and reinforcement design of disk and plate structures with multiple stiffness and eigenfrequency objectives. Comput Struct 1999;72:535–63, URL https://doi.org/10.1016/S0045-7949(98)00326-5.
[30] Olhoff N, Ronholt E, Scheel J. Topology optimization of three-dimensional structures using optimum microstructures. Struct Optim 1998;16:1–18, URL https://doi.org/10.1007/BF01213995.
[31] Rozvany GIN, Bendsoe MP, Kirsch U. Layout optimization of structures. Appl Mech Rev 1995;48(1):41–119, URL https://doi.org/10.1115/1.3005097.
[32] Nadal E, Ródenas JJ, Albelda J, Tur M, Tarancón JE, Fuenmayor FJ. Efficient finite element methodology based on Cartesian grids: Application to structural shape optimization. Abstr Appl Anal 2013;2013:1–19, URL https://doi.org/10.1155/2013/953786.
[33] Dauge M, Düster A, Rank E. Theoretical and numerical investigation of the finite cell method. J Sci Comput 2015;65:1039–64, URL https://doi.org/10.1007/s10915-015-9997-3.
[34] Demmel JW. Applied numerical linear algebra. SIAM; 1997, http://dx.doi.org/10.1137/1.9781611971446.
[35] Wang F, Lazarov BS, Sigmund O. On projection methods, convergence and robust formulations in topology optimization. Struct Multidiscip Optim 2011;43:767–84, URL https://doi.org/10.1007/s00158-010-0602-y.
[36] Sigmund O. Morphology-based black and white filters for topology optimization. Struct Multidiscip Optim 2007;33:401–24, URL https://doi.org/10.1007/s00158-006-0087-x.
[37] Kumar T, Suresh K. A density-and-strain-based K-clustering approach to microstructural topology optimization. Struct Multidiscip Optim 2020;61:1399–415, URL https://doi.org/10.1007/s00158-019-02422-4.
[38] Wu J, Aage N, Westermann R, Sigmund O. Infill optimization for additive manufacturing-approaching bone-like porous structures. IEEE Trans Vis Comput Graphics 2018;24:1127–40, URL https://doi.org/10.1109/TVCG.2017.2655523.
[39] Groen J, Stutz F, Aage N, Bærentzen J, Sigmund O. De-homogenization of optimal multi-scale 3D topologies. Comput Methods Appl Mech Engrg 2020;364:112979. http://dx.doi.org/10.1016/j.cma.2020.112979, URL https://www.sciencedirect.com/science/article/pii/S0045782520301626.
[40] Behrou R, Lotfi R, Carstensen J, Ferrari F, Guest J. Revisiting element removal for density-based structural topology optimization with reintroduction by heaviside projection. Comput Methods Appl Mech Engrg 2021;380:113799. http://dx.doi.org/10.1016/j.cma.2021.113799, URL https://www.sciencedirect.com/science/article/pii/S0045782521001353.
[41] Dedè L, Borden MJ, Hughes TJ. Isogeometric analysis for topology optimization with a phase field model. Arch Comput Methods Eng 2012;19:427–65, URL https://doi.org/10.1007/s11831-012-9075-z.
[42] Chandrasekhar A, Sridhara S, Suresh K. Graded multiscale topology optimization using neural networks. Adv Eng Softw 2023;175:1–10, URL https://doi.org/10.1016/j.advengsoft.2022.103359.
[43] Amstutz S, Novotny AA. Topological optimization of structures subject to Von Mises stress constraints. Struct Multidiscip Optim 2010;41:407–20, URL https://doi.org/10.1007/s00158-009-0425-x.
[44] Gonąlves DC, Lopes JD, Campilho RD, Belinha J. Topology optimization of light structures using the natural neighbour radial point interpolation method. Meccanica 2022;57:659–76, URL https://doi.org/10.1007/s11012-021-01459-4.
[45] Panesar A, Abdi M, Hickman D, Ashcroft I. Strategies for functionally graded lattice structures derived using topology optimisation for additive manufacturing. Addit Manuf 2018;19:81–94, URL https://doi.org/10.1016/j.addma.2017.11.008.